AI Student Impact

AI Impact on Academic Performance, Learning Outcomes, and Burnout Risk: Analytics and Predictive Modeling Dashboard.

Dataset Summary

Students	Features	Completeness
50K	16	100%

Clean dataset with no missing values

Dashboard Modules

- Dataset Exploration
- Advanced Analytics
- Machine Learning Models
- GPA Prediction Tool
- Research Insights

Key Findings

Academic Improvement

• **87.52%** of students improved their GPA after AI-assisted learning.

Average GPA Gain

• Mean improvement: **+0.20 GPA points**

Burnout Risk

Heavy AI users (25+ hours/week) experienced approximately **83% burnout risk.**

Dependency Risk

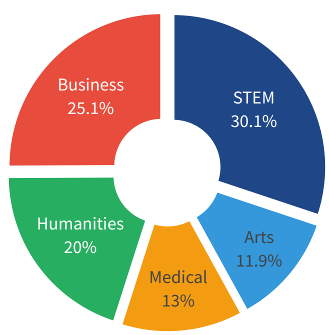
Excessive AI reliance is associated with reduced independent learning and critical thinking.

Dataset Overview & Exploration | Advanced Analytics | Machine Learning Models | Predictions | Insights

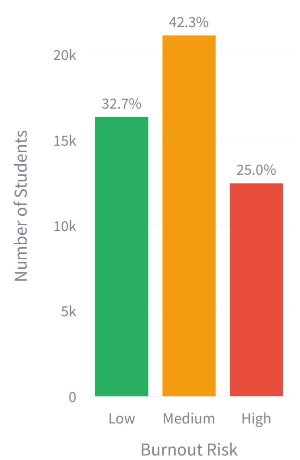
Dataset Overview & Exploration

Avg GPA Imp.	Avg Skill Retention	High Burnout Risk	Avg AI Dependency
+0.203	75.8	25.0%	3.5
87.5% Stud. improved	Std. Dev. ±13.3	12,487 students	Std. Dev. ±1.8
9/10 had a positive GPA change	Retained at ~76%	1 in 4 at risk	35% of the max possible dependency

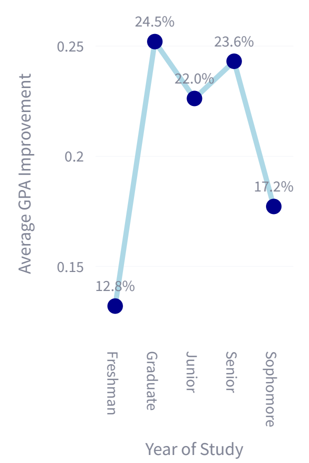
Student Majors: STEM Leads, Arts Trails



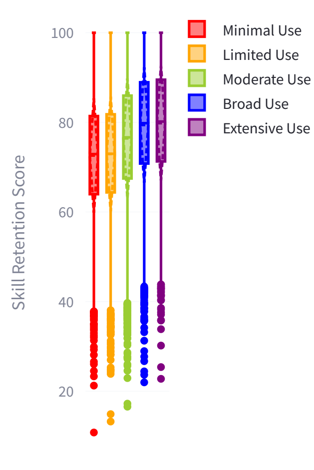
Burnout Risk Levels Across Students



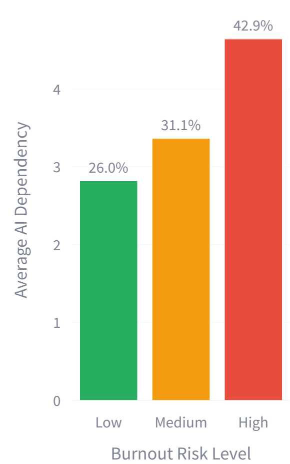
Graduate and Senior Students Show Greatest GPA Gains

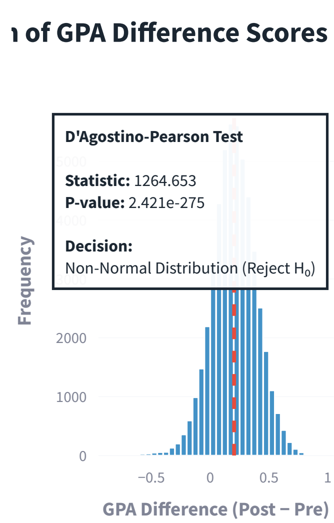


Distribution of Skill Retention Scores by Tool Diversity



AI Dependancy Scores





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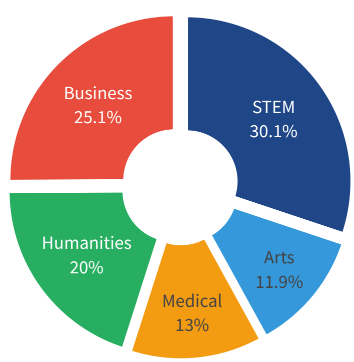

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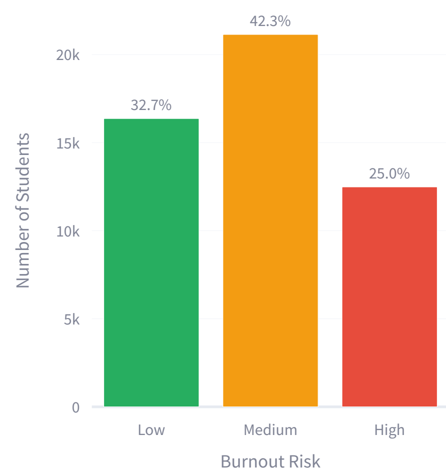

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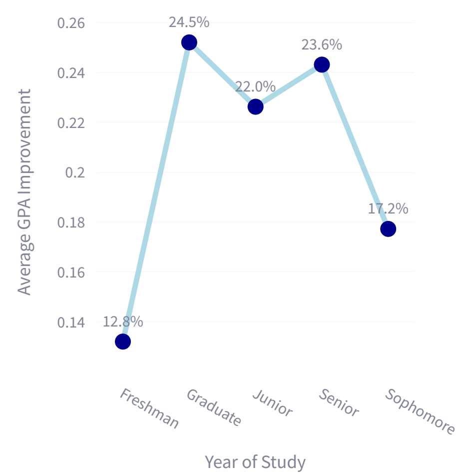

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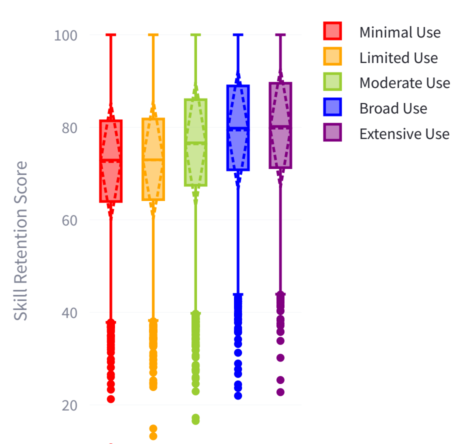
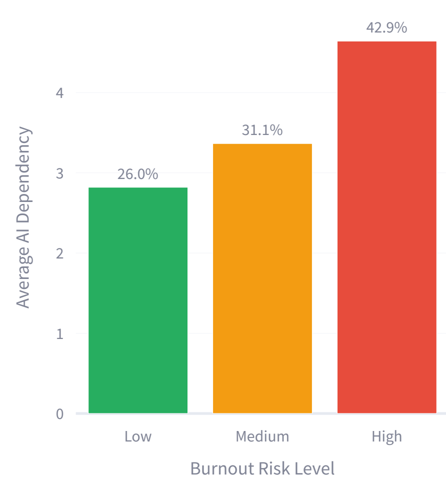
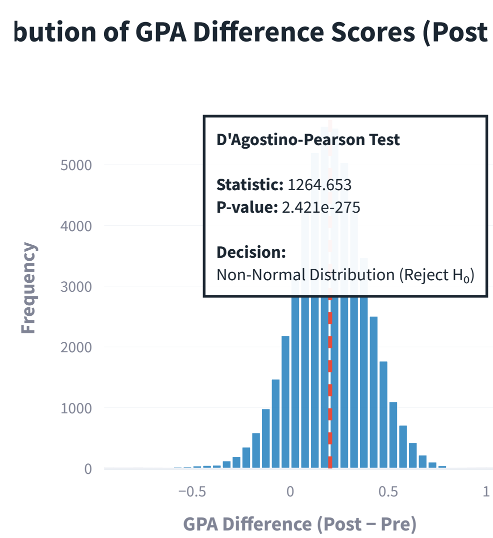


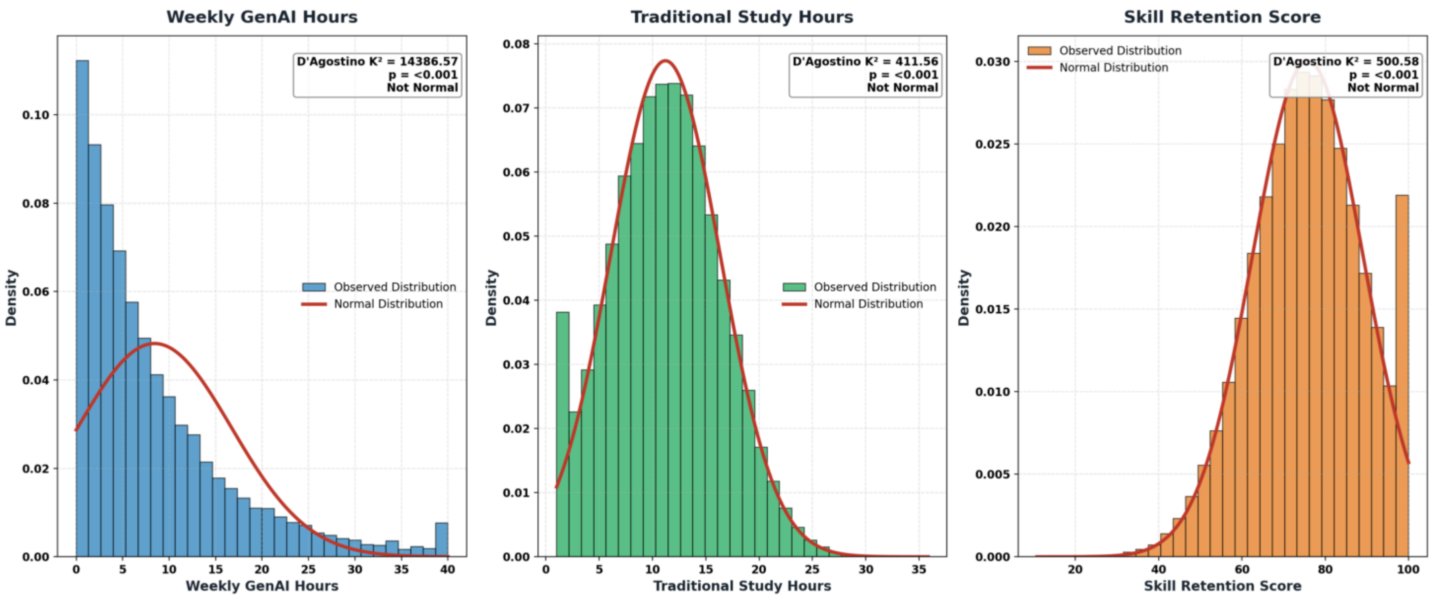
AI Dependancy Scores





1 2 3 4 5
Tool Diversity (Numeric Levels)

Histogram with Fitted Normal Distribution Curves for Continuous Variables



Raw Data Exploration

Filter by Major

All ×

Filter by Burnout Risk

All ×

	Student_ID	Major_Category	Year_of_Study	Pre_Semester_GPA	Weekly_GenAI_Hours	Primary_Use_Case	Prompt_Engineering_Skill	Tool_Diversity	Paic
0	100001	Humanities	Senior	2.418	23.31	Copywriting/Drafting	Beginner	1	
1	100002	Medical	Junior	3.821	1.12	Ideation	Advanced	5	
2	100003	Business	Freshman	3.398	21.26	Summarizing_Reading	Beginner	2	
3	100004	Business	Senior	3.789	1.82	Copywriting/Drafting	Intermediate	4	
4	100005	STEM	Sophomore	3.635	9.29	Debugging/Troubleshooting	Advanced	4	
5	100006	STEM	Junior	3.449	6.5	Debugging/Troubleshooting	Beginner	1	
6	100007	STEM	Freshman	3.622	31.41	Summarizing_Reading	Advanced	5	
7	100008	Arts	Junior	2.746	5.33	Copywriting/Drafting	Intermediate	3	
8	100009	Business	Sophomore	3.42	2	Debugging/Troubleshooting	Beginner	2	
9	100010	Business	Sophomore	3.046	19.99	Debugging/Troubleshooting	Intermediate	2	

AI Student Impact Analytics Dashboard | *Evidence-Based Insights for Stakeholders*

Dataset: 50,000 students | 16 variables | ML models with real-time predictions | Powered by regression & classification frameworks

Recommendations grounded in correlation analysis, PEARSON r*** (p < 0.001), feature importance matrices, and 50,000-student empirical validation.

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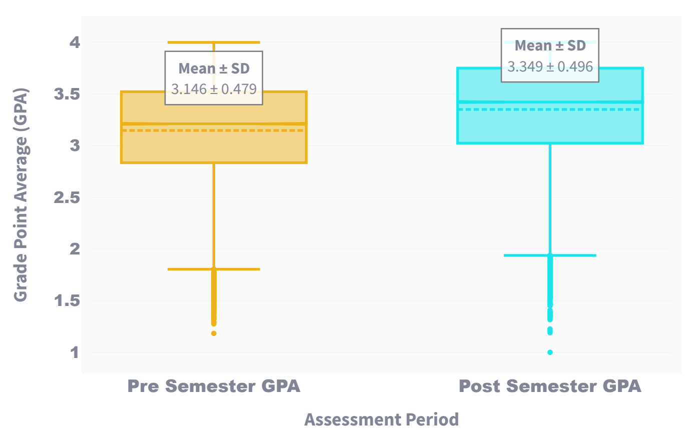
Advanced Analytics

Academic Performance Analysis | AI Usage Patterns & Behavior | Burnout Risk & Mental Health | Skill Retention & Learning Outcomes | Stratified Analysis

GPA Impact Overview

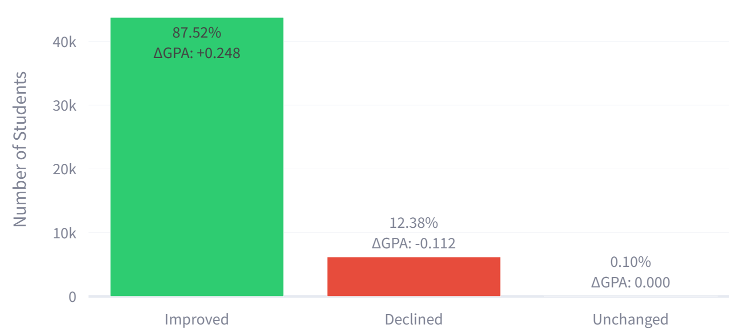
Pre- vs. Post-Semester GPA Distribution



Metric	Pre-Semester GPA	Post-Semester GPA
Mean	3.146	3.349
Std Dev	0.479	0.496
Median	3.210	3.421
Min-Max Range	1.2 - 4.0	1.0 - 4.0
Students Improved	-	87.52%

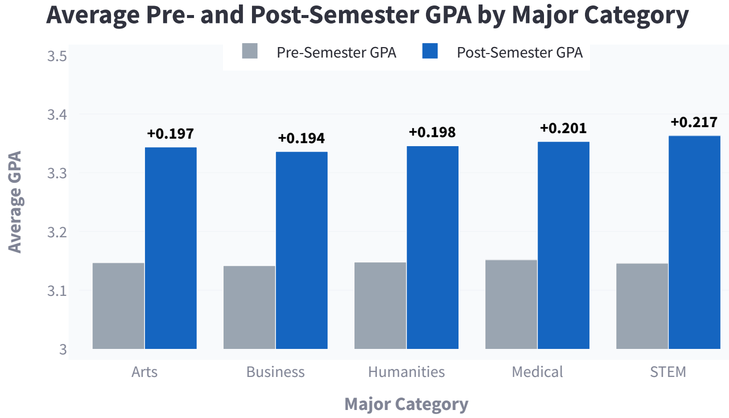
Measure	Value
T-statistic	-242.726
Equivalent F-statistic	58915.856
P-value	< 0.001
Decision	**Mean Difference is Statistically Significant**

Distribution of Student Outcomes

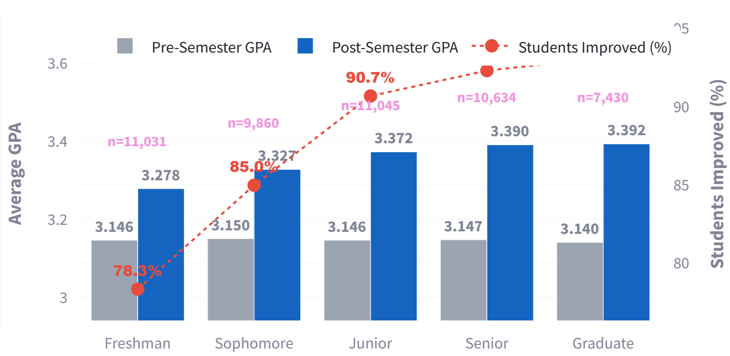


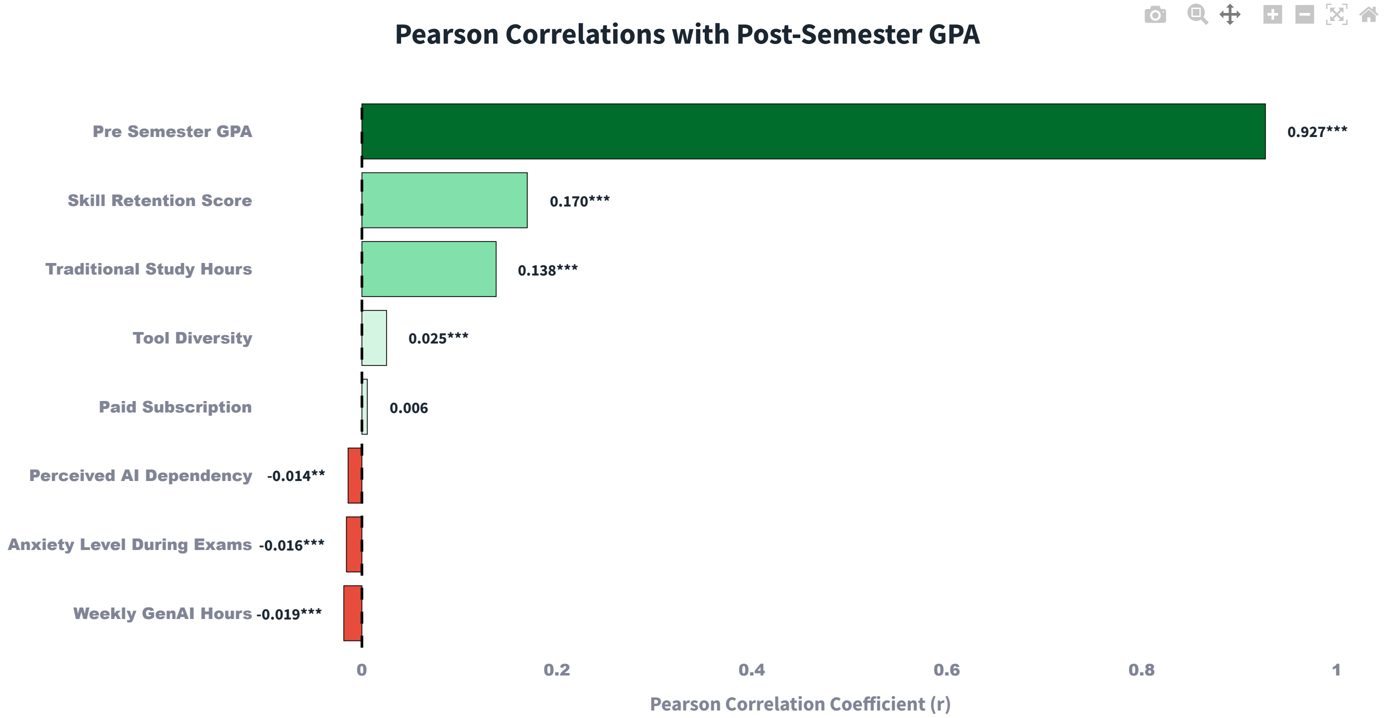
Academic Performance Trends by Major and Year of Study

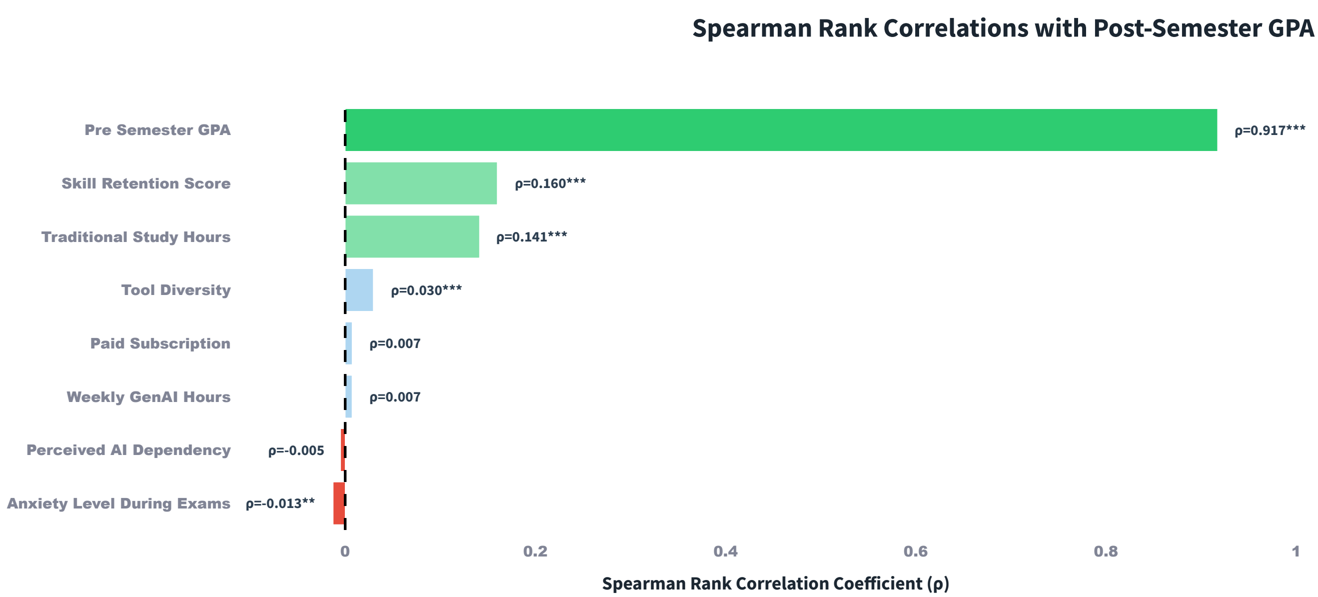
Academic Performance Improved Across All Major Disciplines
Average Pre- and Post-Semester GPA by Major Category



Academic Performance Improves Consistently Across All Years of Study







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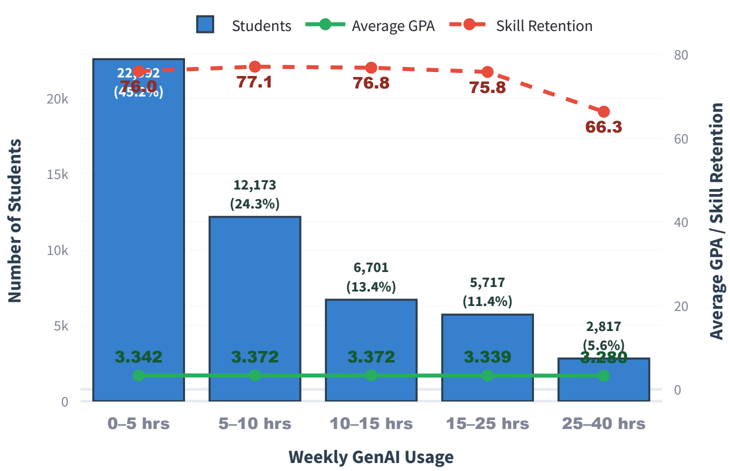
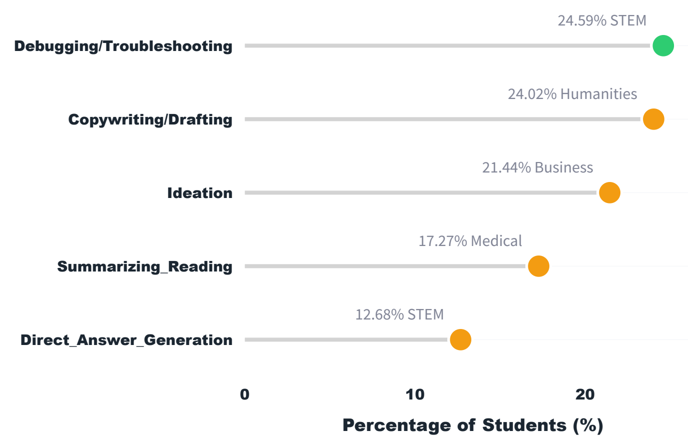
Usage Intensity Distribution

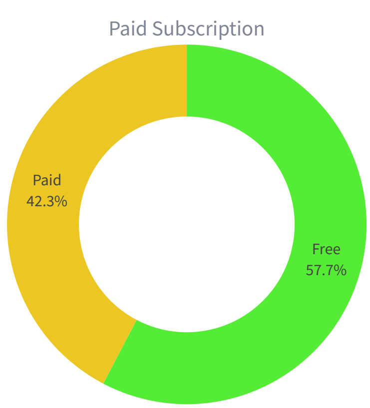
Distribution of Students Across Weekly GenAI Usage Levels



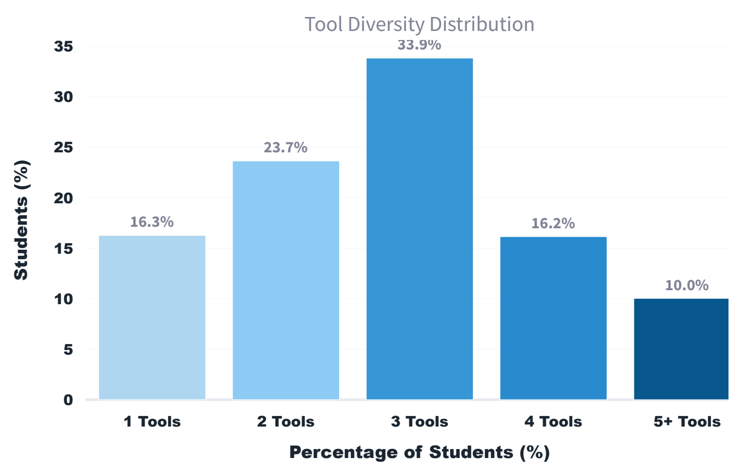
Primary Use Cases of Generative AI



Tool Diversity and Access



Average Tools per Student: 2.80 ± 1.19



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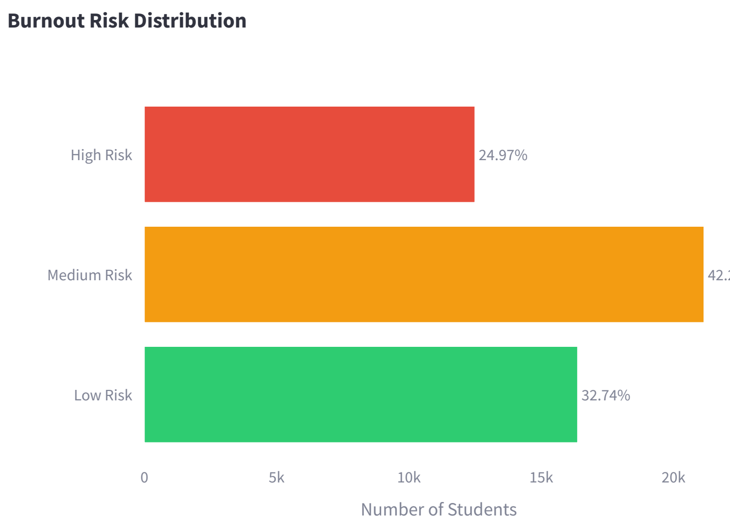
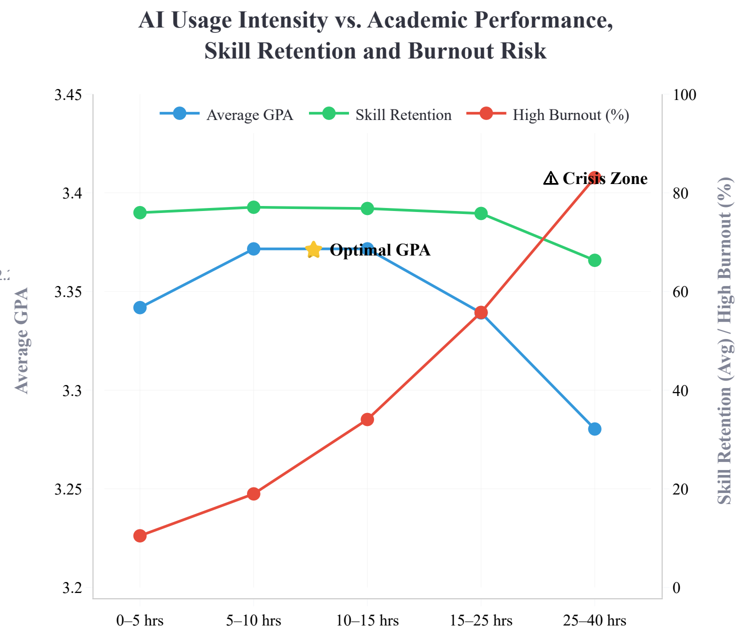
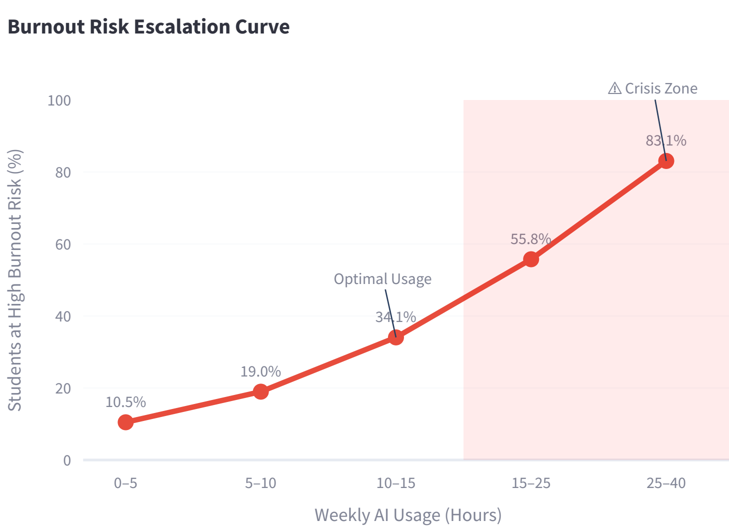
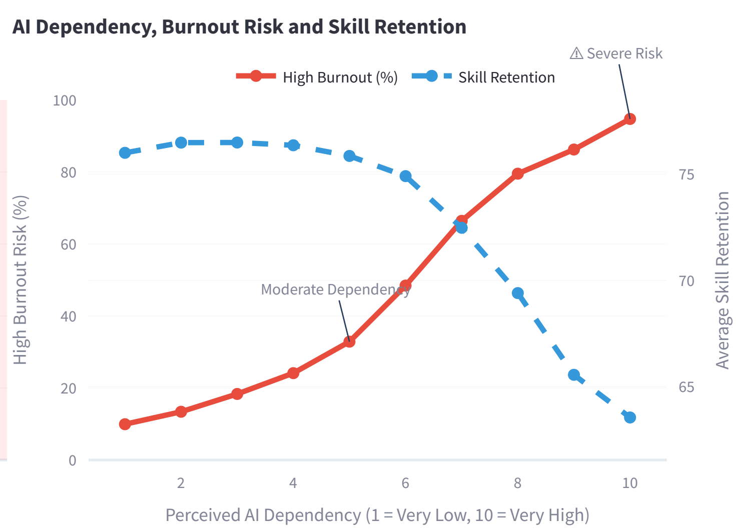
Burnout Risk Distribution



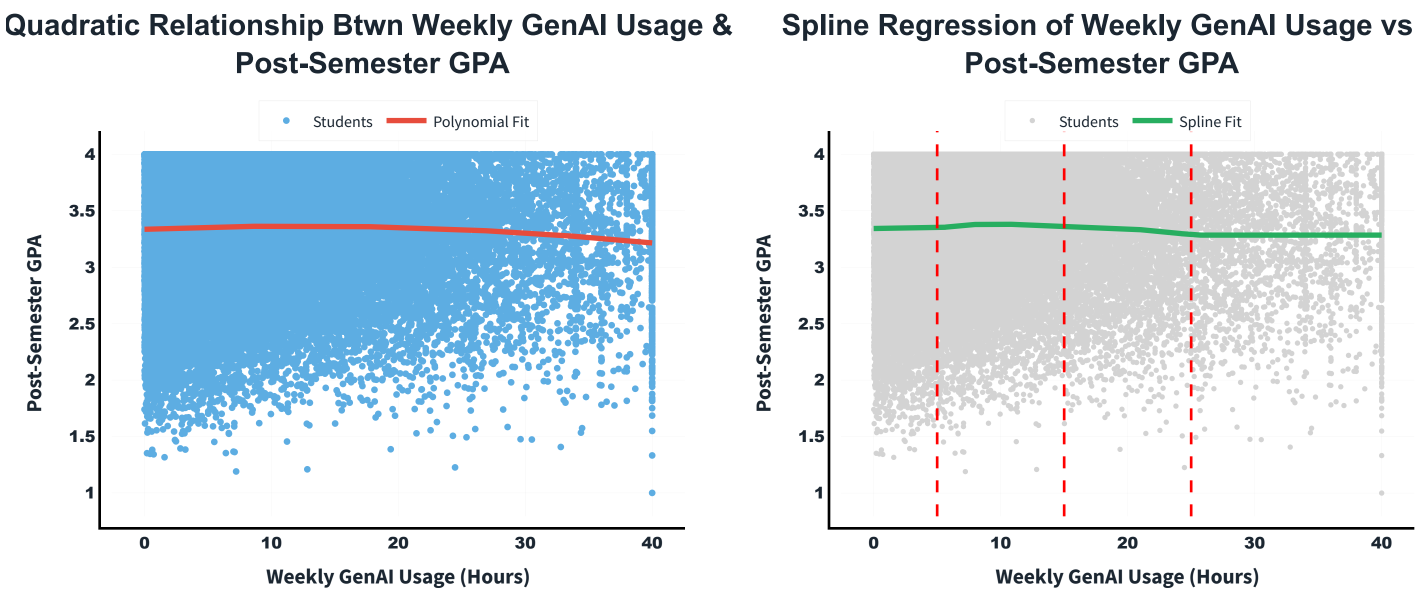


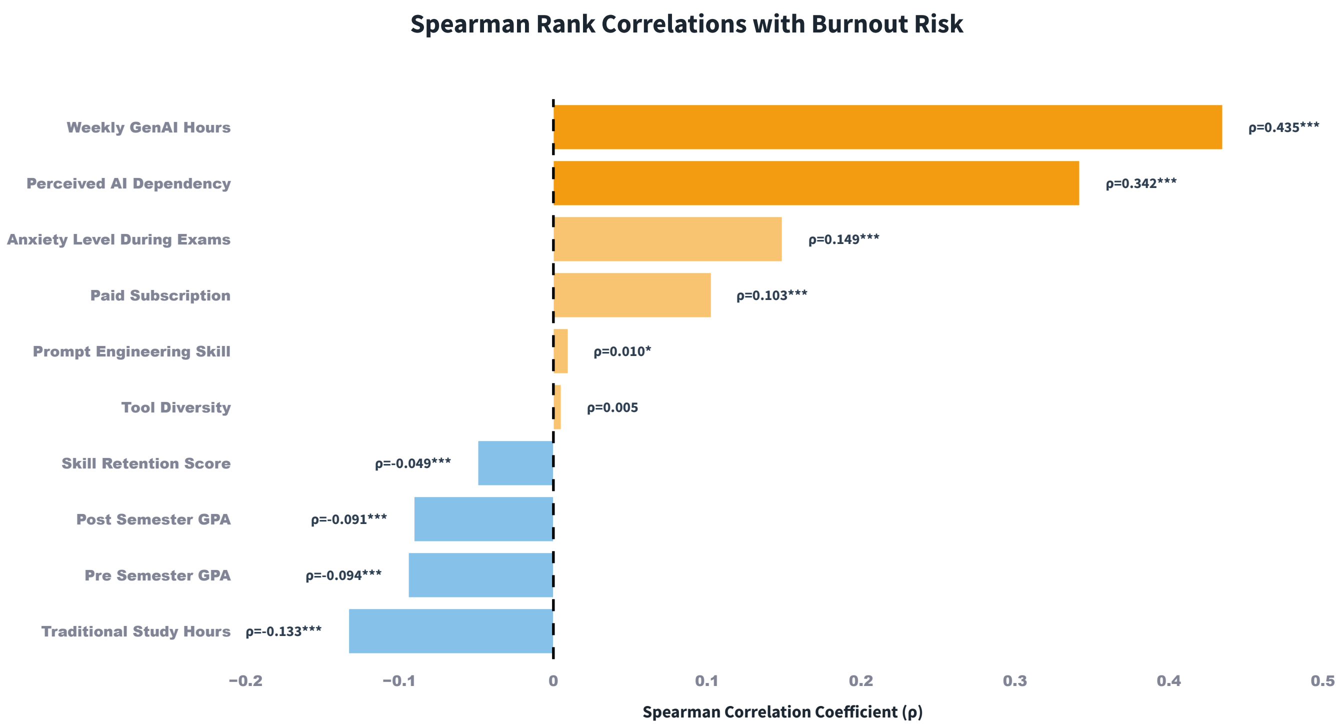
Burnout Risk Escalation Curve





Nonlinear Relationship Between Weekly Generative AI Usage and Academic Performance





Correlation of Variables with Burnout Risk

Variable	Spearman ρ	Strength	Direction	P-value	Significance
Weekly GenAI Hours	+0.435	Moderate	Positive	<0.001	***
Perceived AI Dependency	+0.342	Moderate	Positive	<0.001	***
Anxiety Level During Exams	+0.149	Weak	Positive	<0.001	***
Traditional Study Hours	-0.133	Weak	Negative	<0.001	***
Paid Subscription	+0.103	Weak	Positive	<0.001	***
Pre Semester GPA	-0.094	Negligible	Negative	<0.001	***
Post Semester GPA	-0.091	Negligible	Negative	<0.001	***
Skill Retention Score	-0.049	Negligible	Negative	<0.001	***
Prompt Engineering Skill	+0.010	Negligible	Positive	0.0331	*
Tool Diversity	+0.005	Negligible	Positive	0.2638	

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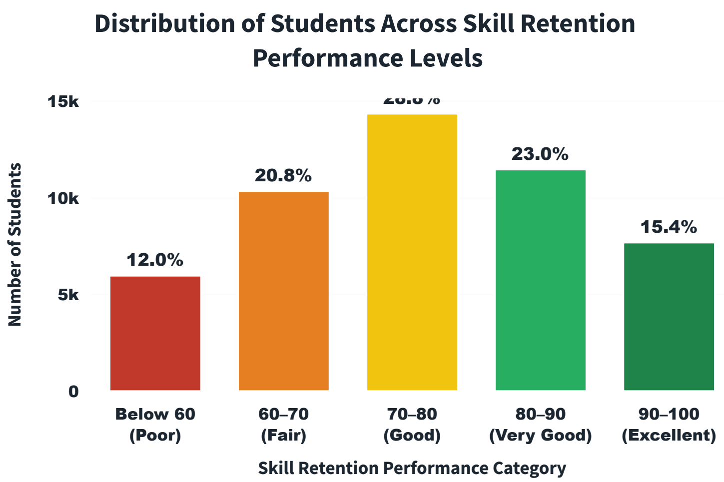
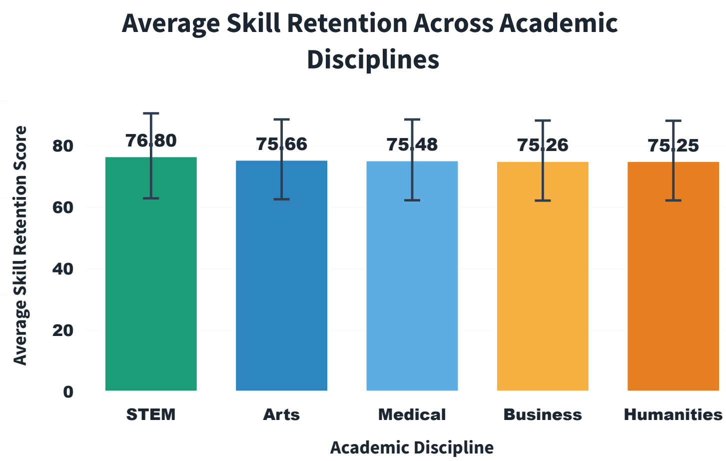
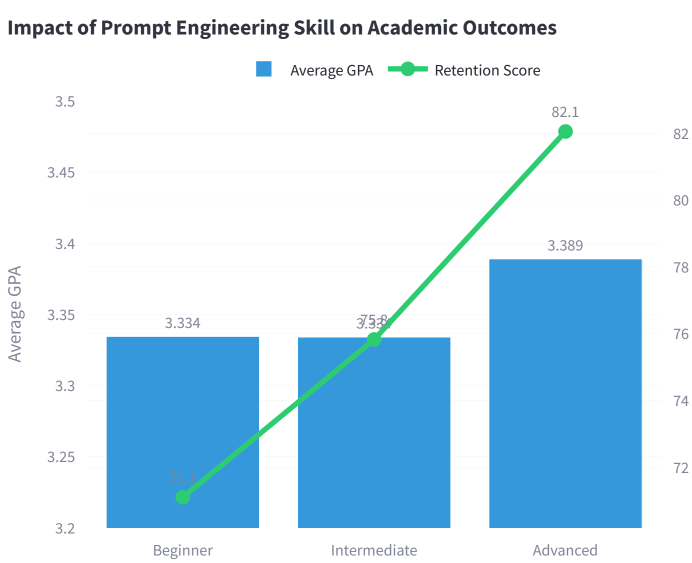
Advanced Analytics

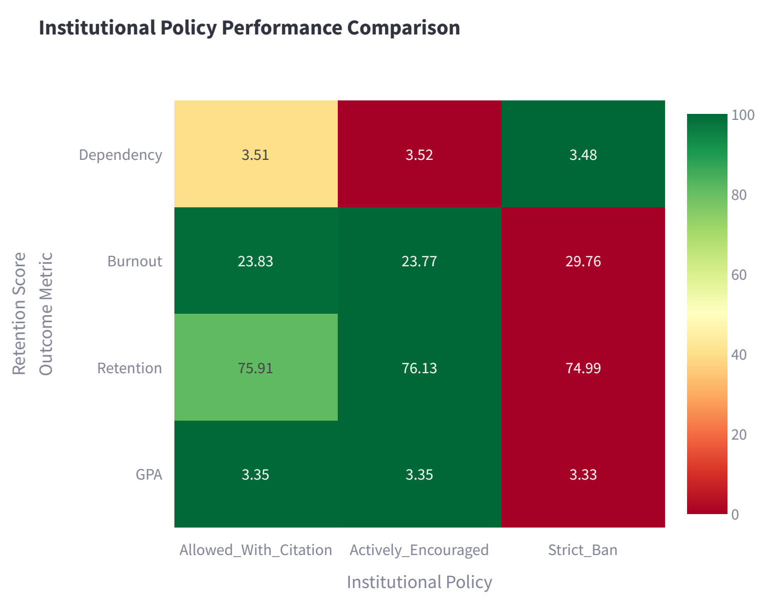
SKILL RETENTION AND LEARNING OUTCOMES





Prompt Engineering Skill Impact





Policy	Mean GPA	Skill Retention	High Burnout (%)	AI Dependency
Allowed_With_Citation	3.35	75.91	23.83	3.5
Actively_Encouraged	3.35	76.13	23.77	3.5
Strict_Ban	3.33	74.99	29.76	3.4

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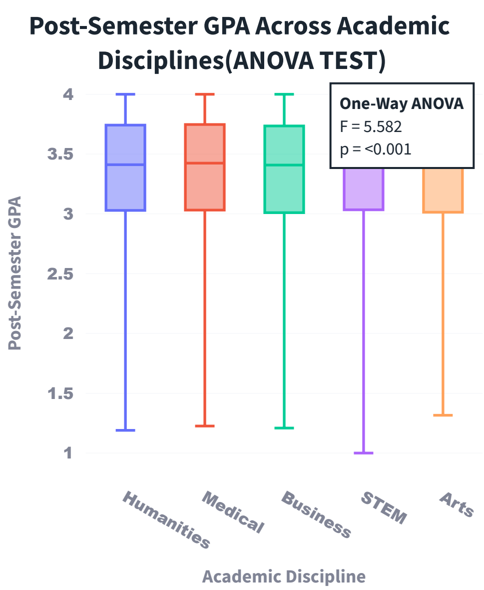
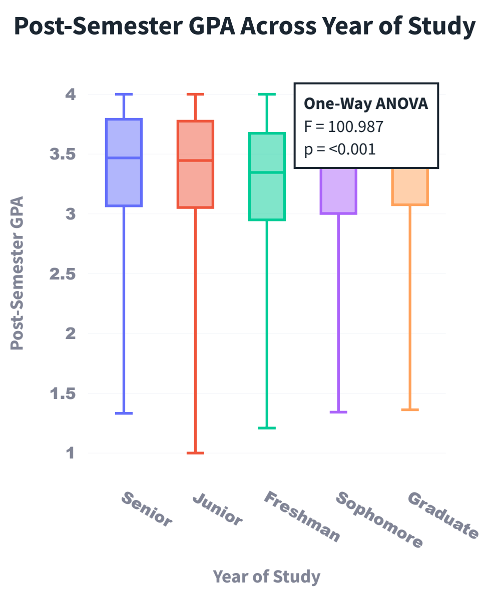
Statistical Testing of Academic Performance Across Student Subgroups

Examination of post-semester GPA differences across academic discipline, year of study, and institutional AI policies using one-way ANOVA with post-hoc Tukey HSD pairwise comparisons.

One-Way ANOVA Analyses

One-way analysis of variance (ANOVA) was conducted to examine whether post-semester GPA differed significantly across student subgroups. This approach tests the null hypothesis that all group means are equal, providing an omnibus test of overall differences. Three distinct stratifications were examined: academic discipline, year of study, and institutional AI policy environment. Statistical significance was evaluated at α = 0.05.





Post-hoc Tukey HSD Pairwise Comparisons

Following significant ANOVA results, post-hoc pairwise comparisons using Tukey's Honestly Significant Difference (HSD) test were performed to identify which specific group pairs differed significantly. This approach controls for family-wise error rate while conducting multiple comparisons. The Tukey HSD test provides both mean differences and 95% confidence intervals for each pairwise contrast, with adjusted p-values indicating statistical significance at the 0.05 level.

Post-Semester GPA Across Academic Disciplines

Post-hoc Tukey HSD Pairwise Test

Category 1	Category 2	Mean Difference	Adjusted p-value	Lower CI	Upper CI	Significant
Arts	Business	-0.0076	0.8672	-0.0289	0.0137	☐
Arts	Humanities	0.0021	0.9989	-0.02	0.0243	☐
Arts	Medical	0.0095	0.8248	-0.0148	0.0338	☐
Arts	STEM	0.0195	0.0774	-0.0013	0.0402	☐
Business	Humanities	0.0097	0.5849	-0.0084	0.0279	☐
Business	Medical	0.0171	0.1607	-0.0036	0.0378	☐
Business	STEM	0.0271	0.0001	0.0107	0.0434	☑
Humanities	Medical	0.0073	0.8858	-0.0142	0.0289	☐
Humanities	STEM	0.0173	0.0525	-0.0001	0.0348	☐
Medical	STEM	0.01	0.6559	-0.0101	0.0301	☐

Post-Semester GPA Across Year of Study

Post-hoc Tukey HSD Pairwise Test

Year 1	Year 2	Mean Difference	Adjusted p-value	Lower CI	Upper CI	Significant
Freshman	Graduate	0.1146	0	0.0944	0.1348	☑
Freshman	Junior	0.0941	0	0.076	0.1123	☑
Freshman	Senior	0.1124	0	0.0941	0.1307	☑
Freshman	Sophomore	0.0493	0	0.0306	0.0679	☑
Graduate	Junior	-0.0204	0.046	-0.0406	-0.0002	☑
Graduate	Senior	-0.0022	0.9983	-0.0226	0.0181	☐
Graduate	Sophomore	-0.0653	0	-0.086	-0.0446	☑
Junior	Senior	0.0182	0.0517	-0.0001	0.0365	☐
Junior	Sophomore	-0.0449	0	-0.0635	-0.0262	☑
Senior	Sophomore	-0.0631	0	-0.0819	-0.0442	☑

Post-Semester GPA Across Institutional Policies

Post-hoc Tukey HSD Pairwise Test

Policy 1	Policy 2	Mean Difference	Adjusted p-value	Lower CI	Upper CI	Significant
Encouraged	Citation	0	1	-0.012	0.0119	☐
Encouraged	Strict Ban	-0.0207	0.0038	-0.0358	-0.0056	☑
Citation	Strict Ban	-0.0206	0.0014	-0.0345	-0.0068	☑

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Machine Learning Models

Select Prediction Task

- [x] GPA Prediction
- [ ] Burnout Classification
- [ ] Skill Retention

Predicting Post-Semester GPA

Advanced machine learning models to predict student GPA based on study patterns and AI tool usage.

Model Performance Overview

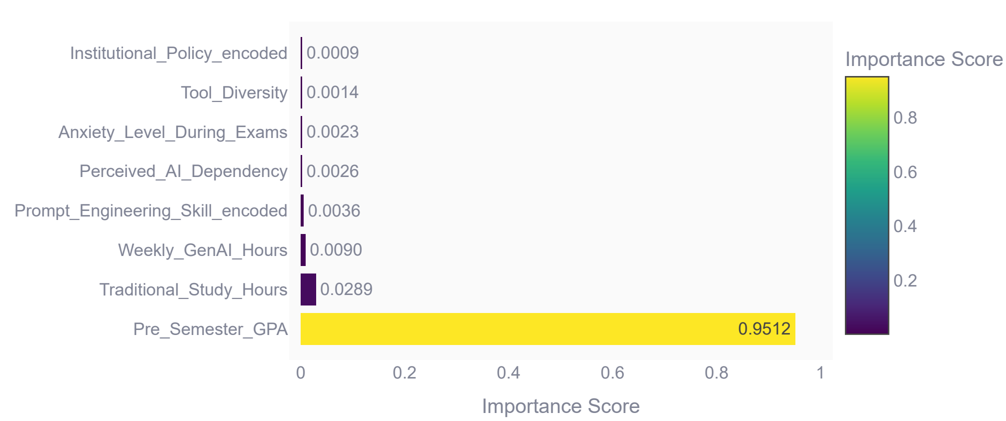
BEST MODEL	R^2 SCORE	RMSE	MAE
Gradient	**0.887**	**0.165**	**0.127**
	Explains 88.7%	GPA Points	Avg Error

Model Comparison Summary

Model	R^2 Score	RMSE	MAE	MAPE (%)
Gradient Boosting	0.8871	0.1651	0.1273	3.9853
Random Forest	0.8845	0.167	0.1284	4.0231
Linear Regression	0.8819	0.1688	0.1314	4.0938

Feature Importance Analysis

Which Factors Drive GPA Most?



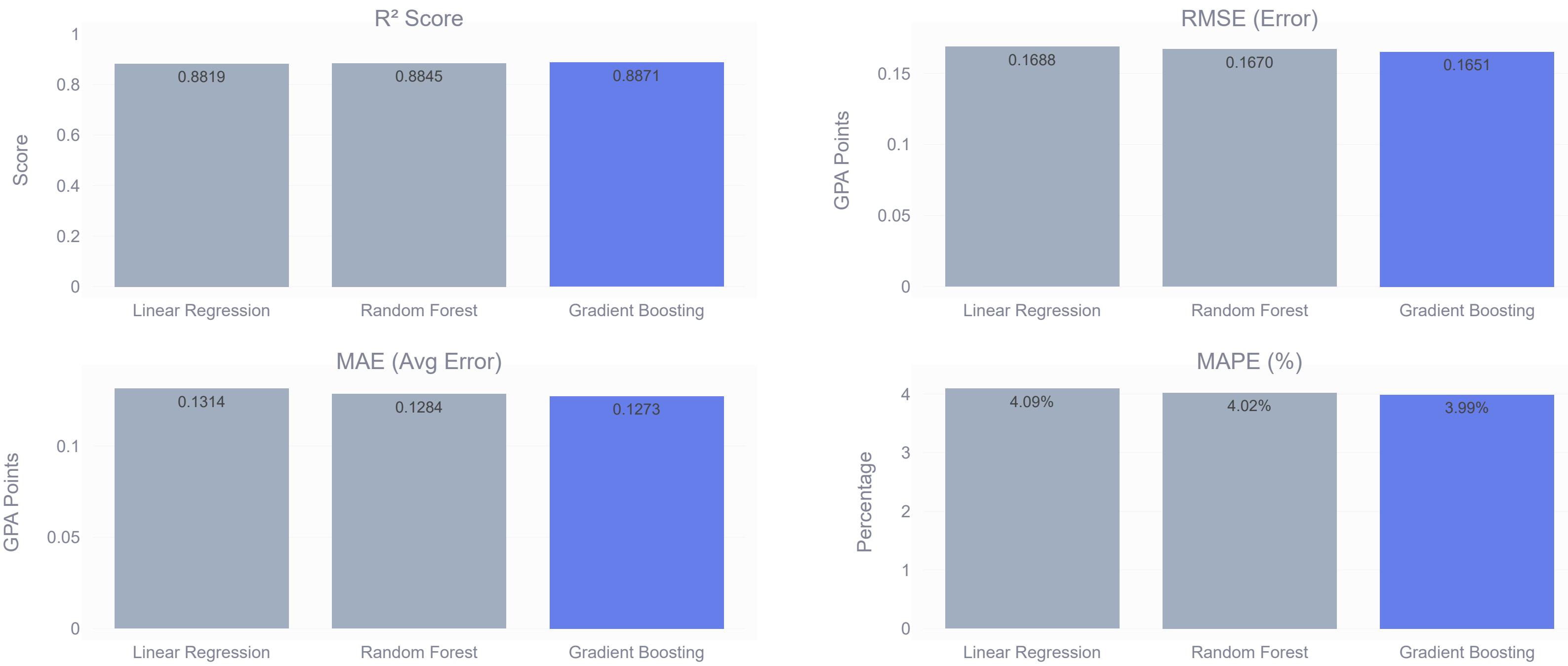
Top 3 Predictors:

1. **Pre_Semester_GPA**
   95.1% importance
2. **Traditional_Study_Hours**
   2.9% importance
3. **Weekly_GenAI_Hours**
   0.9% importance

Key Insight: Pre-semester performance and study methods are the strongest predictors of post-semester success.

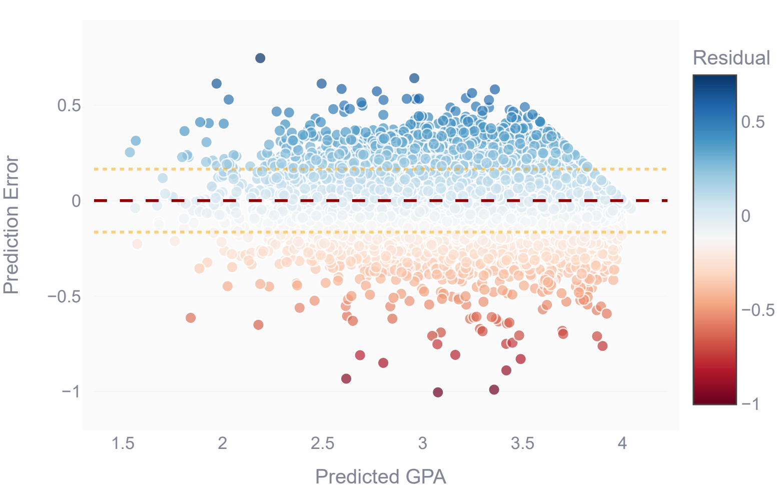
Detailed Metrics Comparison

Performance Metrics Comparison

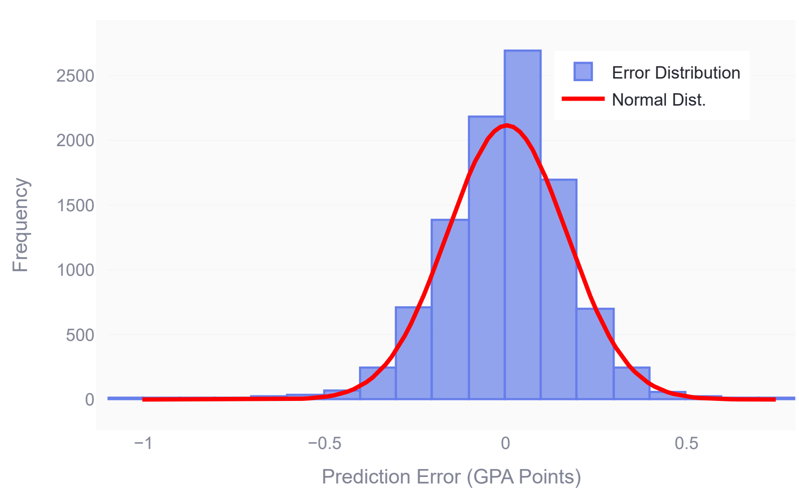


Prediction Accuracy Analysis

Residuals vs Predicted GPA



Error Distribution



Error Statistics

Median Error	Max Error	Within ±0.5	Within ±1.0
0.103 GPA	1.004 GPA	99.2%	100.0%

Model Details

Linear Regression | Random Forest | Gradient Boosting

⭐ ⭐ ⭐ ⭐ Strong

Performance Metrics:

- **R^2 Score:** 0.8819
- **RMSE:** 0.1688 GPA
- **MAE:** 0.1314 GPA
- **MAPE:** 4.09%

What This Means:

- Explains 88.2% of variation
- Avg error: ±0.131 GPA
- Worst case: ±0.169 GPA
- Error is 4.1% of actual

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Machine Learning Models

Select Prediction Task

- ○ GPA Prediction
- ● Burnout Classification
- ○ Skill Retention

Predicting Burnout Risk

Classification models to identify students at risk of burnout based on AI tool usage patterns and stress indicators.

Model Performance Overview

BEST MODEL
Logistic

ACCURACY
0.514
51.4%

F1 SCORE
0.514
Weighted

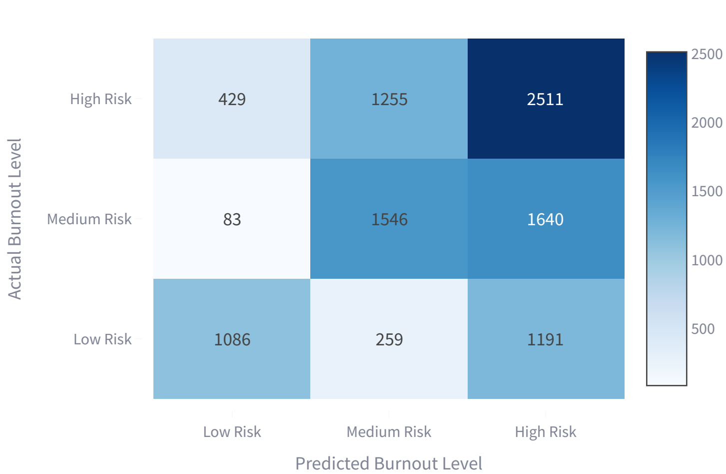
PRECISION
0.535
Avg

Model Comparison

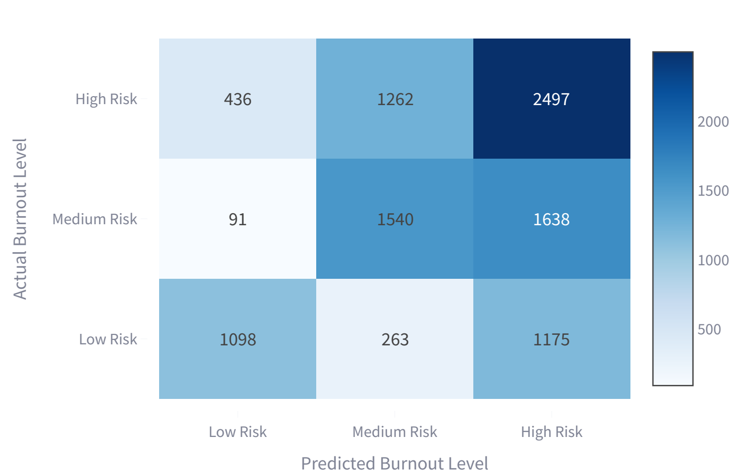
Model	Accuracy	F1 Score	Precision	Recall
Logistic Regression	0.5143	0.5138	0.5347	0.514
Random Forest	0.5135	0.5132	0.5329	0.513

Classification Performance

Logistic Regression - Confusion Matrix

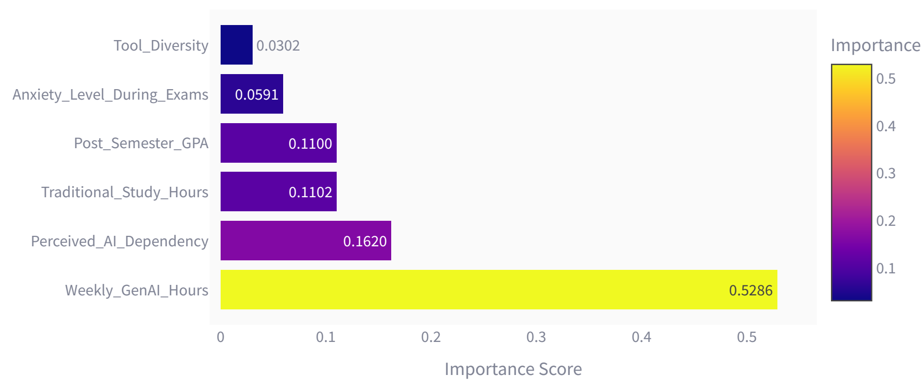


Random Forest - Confusion Matrix



Feature Importance

Which Factors Predict Burnout Risk?



Top 3 Predictors:

1. **Weekly_GenAI_Hours**
   52.9% importance
2. **Perceived_AI_Dependency**
   16.2% importance
3. **Traditional_Study_Hours**
   11.0% importance

Key Insight: Weekly_GenAI_Hours and Perceived_AI_Dependency are the strongest predictors of burnout risk.

Performance Metrics Comparison

Classification Metrics Comparison

Model Details

Logistic Regression | Random Forest

Classification Metrics:

- **Accuracy:** 0.5143 (51.43%)
- **F1 Score:** 0.5138
- **Precision:** 0.5347
- **Recall:** 0.5143

What This Means:

- Correctly classifies 51.4% of students
- F1 balances precision and recall
- Precision: How many identified risks are real
- Recall: How many actual risks are caught

Dataset Overview & Exploration | Advanced Analytics | Machine Learning Models | Predictions | Insights

Machine Learning Models

Select Prediction Task

○ GPA Prediction ○ Burnout Classification ● Skill Retention

Predicting Skill Retention Score

Regression models to predict how well students will retain learned skills based on learning patterns and demographics.

Model Performance Overview

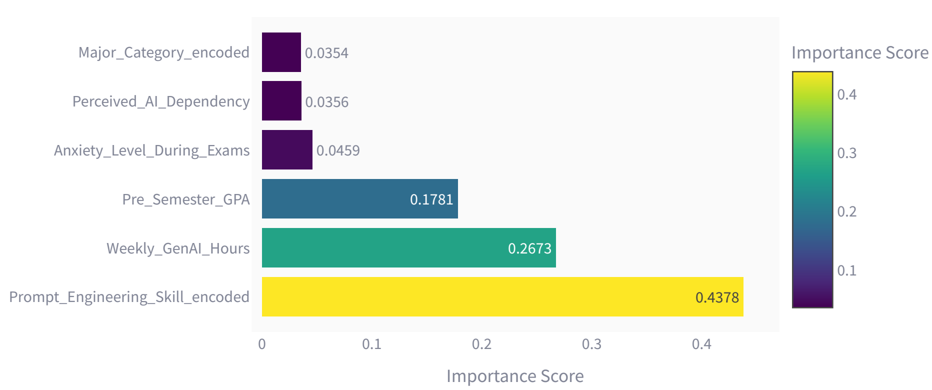
BEST MODEL	R^2 SCORE	RMSE	MAE
Gradient	**0.153**	**12.219**	**9.875**
	Explains 15.3%	Retention %	Avg Error

Model Comparison Summary

Model	R^2 Score	RMSE	MAE	MAPE (%)
Gradient Boosting	0.1531	12.2186	9.875	14.06
Random Forest	0.1488	12.2496	9.8973	14.09
Linear Regression	0.0563	12.8979	10.4509	14.94

Feature Importance Analysis

What Drives Skill Retention?



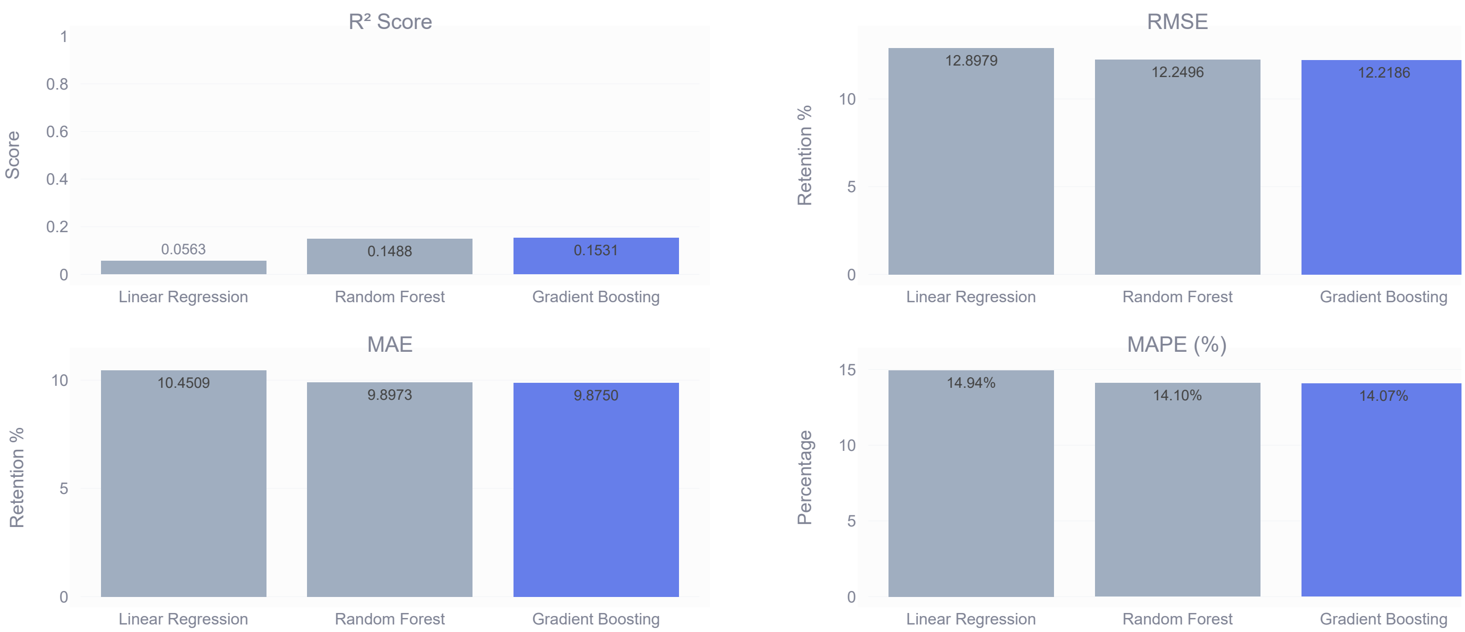
Top 3 Factors:

1. **Prompt_Engineering_Skill_encoded**
   43.8%
2. **Weekly_GenAI_Hours**
   26.7%
3. **Pre_Semester_GPA**
   17.8%

Key Insight: Prompt_Engineering_Skill_encoded and Weekly_GenAI_Hours are the strongest predictors of burnout risk.

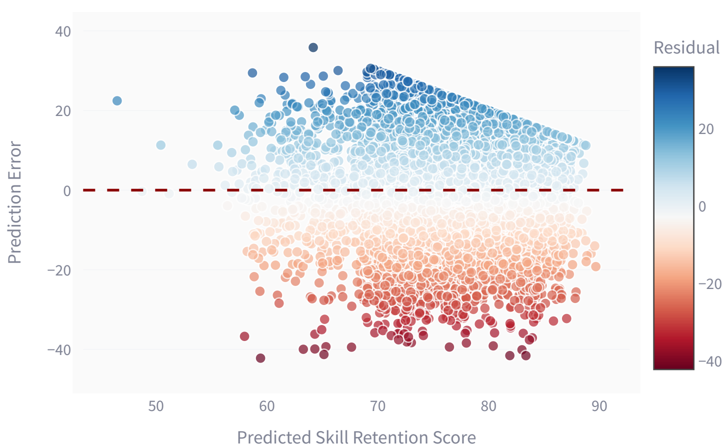
Detailed Metrics Comparison

Performance Metrics Comparison

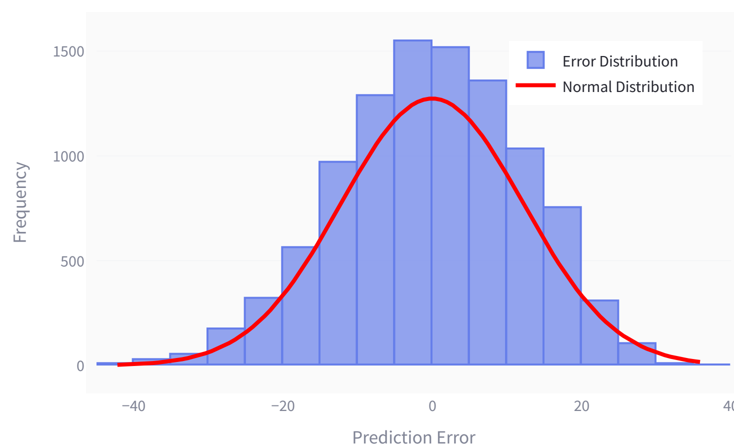


Prediction Accuracy Analysis

Residuals vs Predicted Retention



Error Distribution



Error Statistics

Median Error

8.541

Max Error

42.197

Within ±5%

30.6%

Within ±10%

57.0%

Model Details

Linear Regression | Random Forest | Gradient Boosting

⭐ ⭐ Moderate

Performance Metrics:

- **R^2 Score:** 0.0563
- **RMSE:** 12.8979
- **MAE:** 10.4509
- **MAPE:** 14.94%

Interpretation:

- Explains 5.6% of retention variation
- Average prediction error: ±10.45%
- Relative error: 14.9%
- Model consistency: 85.1%

AI Student Impact Analytics Dashboard | *Evidence-Based Insights for Stakeholders*

Dataset: 50,000 students | 16 variables | ML models with real-time predictions | Powered by regression & classification frameworks

Recommendations grounded in correlation analysis, PEARSON r*** ($p < 0.001$), feature importance matrices, and 50,000-student empirical validation.

Dataset Overview & Exploration | Advanced Analytics | Machine Learning Models | Predictions | Insights

Make Predictions

Select Prediction Type

- (•) Predict Student GPA
- ( ) Assess Burnout Risk
- ( ) Estimate Skill Retention

Student GPA Prediction Model

Enter student details to predict post-semester GPA performance.

Student Information

Pre-Semester GPA

2.50

1.00 — 4.00

AI Dependency Level (1-10)

3

1 — 10

Study Patterns

Traditional Study Hours/Week

12.00

0.00 — 40.00

Weekly GenAI Hours

8.00

0.00 — 40.00

Number of AI Tools Used

2

1 — 5

Exam Anxiety Level (1-10)

3

1 — 10

Skills & Policies

Prompt Engineering Skill Level

Beginner

Institutional AI Policy

Allowed_With_Citation

Generate GPA Prediction

Prediction Results

Comprehensive analysis of predicted post-semester academic performance based on your current study habits, AI tool usage, and academic metrics. This forecast uses machine learning to estimate your likely GPA trajectory and highlight key factors that may impact your success.

PREDICTED POST-SEMESTER GPA

2.81

Your estimated GPA at the end of this semester based on current trajectory

">

RISK STATUS

High Risk

Assessment of academic performance sustainability

MODEL CONFIDENCE

88.7%

Statistical reliability of this forecast based on model training data

Detailed Analysis

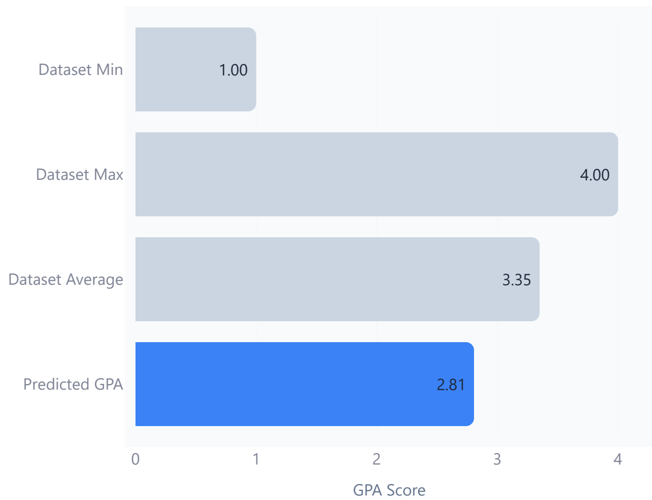
This section provides deeper insights into your predicted academic performance, comparing your trajectory against historical dataset benchmarks while also visualizing the confidence interval of the prediction model. It highlights not only where you currently stand but also the degree of certainty behind the forecast, offering a clearer perspective on potential outcomes and areas of improvement.

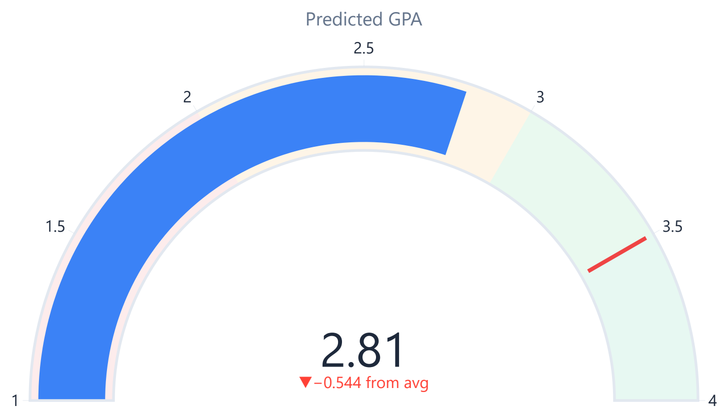
Performance Comparison

Your predicted GPA benchmarked against the distribution from the full dataset. This shows where your forecast falls relative to average, maximum, and minimum performance in our training cohort.

Prediction Confidence Range

Gauge showing your predicted GPA relative to the 1.0–4.0 scale. The colored zones indicate risk levels: red (below 2.0), orange (2.0–3.0), green (3.0–3.5), and optimal (above 3.5). The red threshold marks 3.5 as a strong GPA target.





Key Insights & Recommendations

Based on your input data, the model has identified specific patterns that may impact your academic success. Review the insights below to understand key factors in your prediction, then consider the actionable recommendations tailored to your situation.

Key Insights

Your study patterns show balance across multiple dimensions: reasonable AI usage, adequate study time, and appropriate tool selection. This suggests you're making intentional, thoughtful choices about your learning approach.

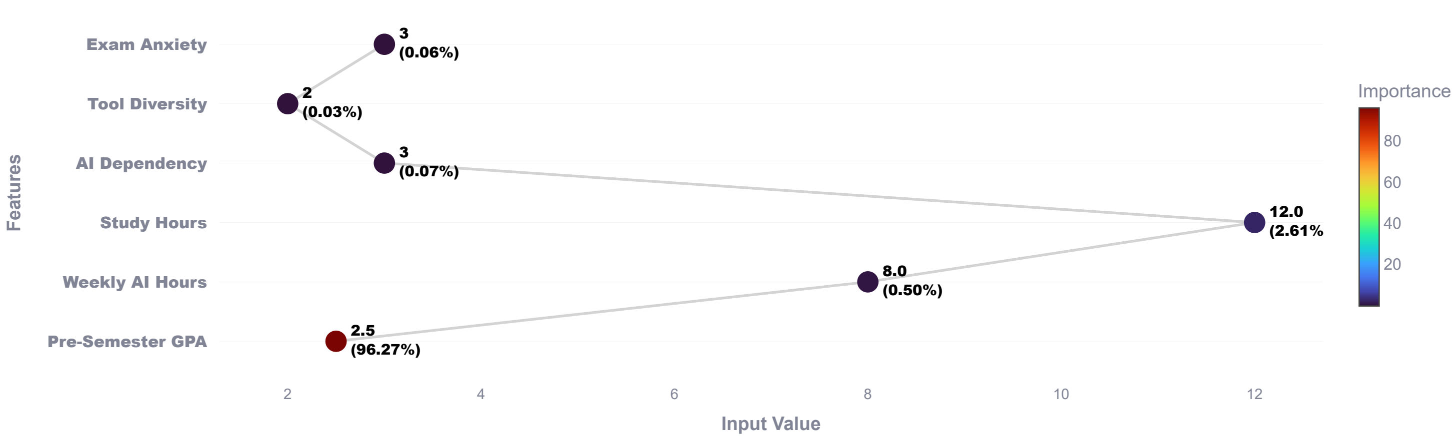
Recommendations

Maintain your current habits and continue monitoring your progress. Regular self-assessment will help you catch any emerging issues early and adjust as coursework demands increase.

What Drives Your GPA?

Analysis of which factors most strongly influence your predicted post-semester GPA. Features are ranked by model importance—darker red indicates stronger predictive power for academic performance.

Student Profile: Feature Values (colored by importance)



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Make Predictions

Select Prediction Type

- ○ Predict Student GPA
- ● Assess Burnout Risk
- ○ Estimate Skill Retention

Student Burnout Risk Assessment

Evaluation of academic burnout risk based on workload intensity, stress levels, and engagement patterns with AI tools. Burnout is a serious concern affecting student wellbeing and academic outcomes. This assessment helps identify risk factors early so you can take preventive action.

Burnout Risk Factors

Input your current situation across multiple dimensions to assess your burnout risk.

Weekly GenAI Hours: 8 (range 0–40)

Exam Anxiety Level (1-10): 4 (range 1–10)

Traditional Study Hours/Week: 12 (range 0–40)

AI Dependency Level (1-10): 3 (range 1–10)

Number of AI Tools Used: 3 (range 1–5)

Current Semester GPA: 3.30 (range 1.00–4.00)

Assess Burnout Risk

Your Burnout Risk Assessment Results

Based on your input data and advanced predictive modeling, here is your personalized burnout risk evaluation. This analysis not only highlights your current risk level but also uncovers underlying drivers such as workload balance, study efficiency, and AI tool dependency. Alongside the forecast, you'll receive actionable recommendations designed to strengthen resilience, optimize learning strategies, and support long-term academic success.

MODEL CONFIDENCE	BURNOUT RISK LEVEL	PEAK PROBABILITY
51.3%	**High Risk**	**49.4%**
Statistical reliability of prediction	Risk Assessment	Strongest prediction indicator

Burnout Risk Analysis

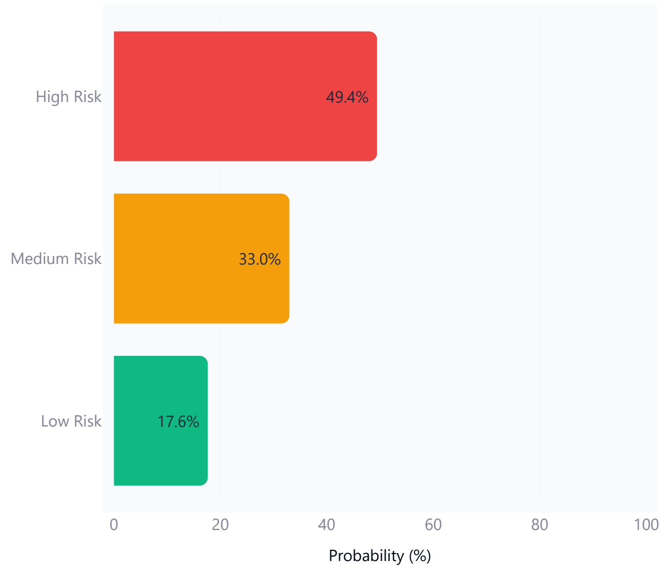
Two perspectives on your predicted burnout risk: your individual probability distribution (left) and how your profile compares to the average student (right).

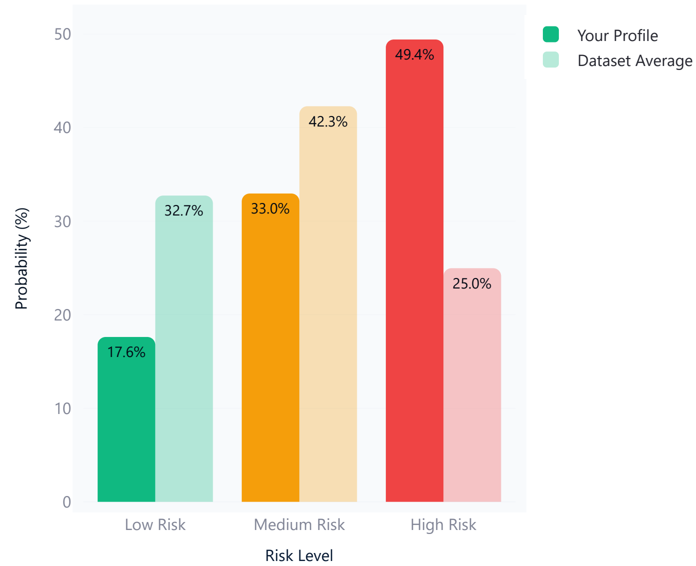
Risk Probability Distribution

Visually presents predicted probabilities across all defined risk levels, displaying separate horizontal bars for high, medium, and low risk categories with their respective probabilities (%) values.

Your Risk vs Dataset Average

Comparison of predicted risk probabilities against the average risk distribution of the dataset, showing side-by-side bars for low, medium, and high risk categories with percentage values."





Personalized Recommendations

Based on your risk level and specific input factors, here are targeted recommendations to help prevent or reduce burnout and improve your overall wellbeing.

💭 Burnout Insights

Your burnout risk factors appear isolated rather than compounded. Strategic intervention in one area could reduce overall risk.

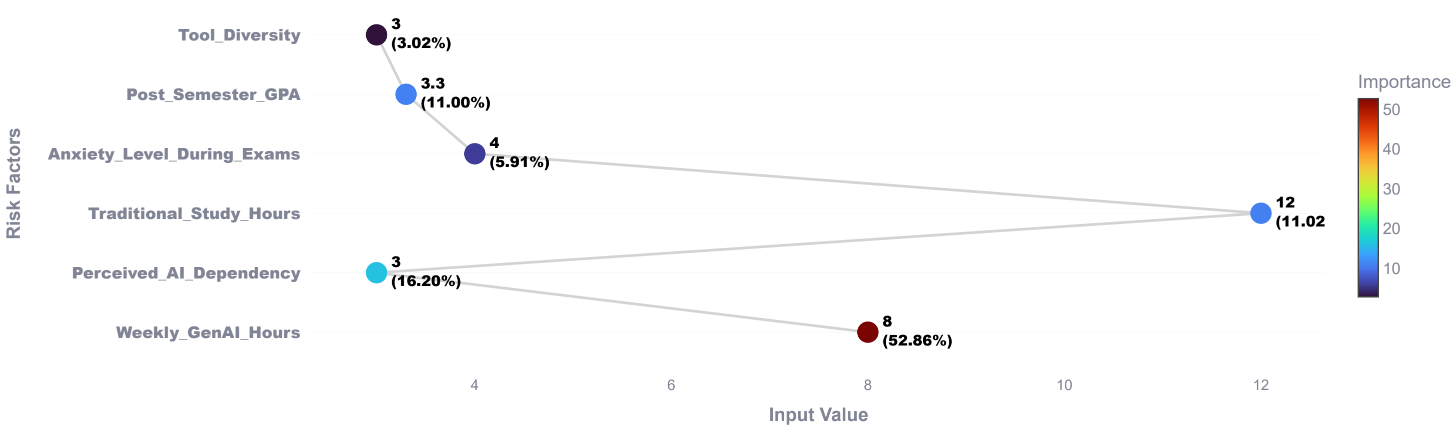
Action Steps

🔴 **URGENT ACTION REQUIRED:** Consider scheduling a meeting with an academic advisor or campus counselor as soon as possible. High burnout risk requires professional support to develop a recovery strategy.

What Predicts Student Burnout?

Key factors influencing burnout risk based on student behavior and emotional indicators. Darker orange markers indicate stronger predictive power—these are the areas to monitor and support.

Burnout Risk Factors: Feature Importance



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Make Predictions

Select Prediction Type

○ Predict Student GPA ○ Assess Burnout Risk ● Estimate Skill Retention

Skill Retention Prediction

Forecast how well you will retain and master skills learned this semester based on your learning profile, engagement with AI tools, and academic foundation.

Your Learning Profile

Enter details on learning, AI skills, stress, and academics for a personalized retention forecast.

Weekly GenAI Hours: 8 (0 – 40)

AI Dependency Level (1-10): 3 (1 – 10)

Prompt Engineering Skill Level: Beginner

Exam Anxiety (1-10): 3 (1 – 10)

Pre-Semester GPA: 2.17 (1.00 – 4.00)

Major Category: STEM

Predict Skill Retention

Your Skill Retention Forecast

Personalized prediction of how well you will retain and apply the foundational knowledge and skills learned this semester, with deeper insights into the specific factors that support or limit your ability to retain knowledge. This forecast evaluates how your study patterns, AI tool usage, anxiety levels, and prompt engineering proficiency collectively influence your long-term skill retention—recognizing that retaining core competencies is just as important as achieving strong grades.

PREDICTED RETENTION SCORE	RETENTION LEVEL	MODEL CONFIDENCE
80.3%	Excellent	15.3%
Expected skill retention at semester end	Performance category	Prediction reliability (R^2 score)

Retention Analysis

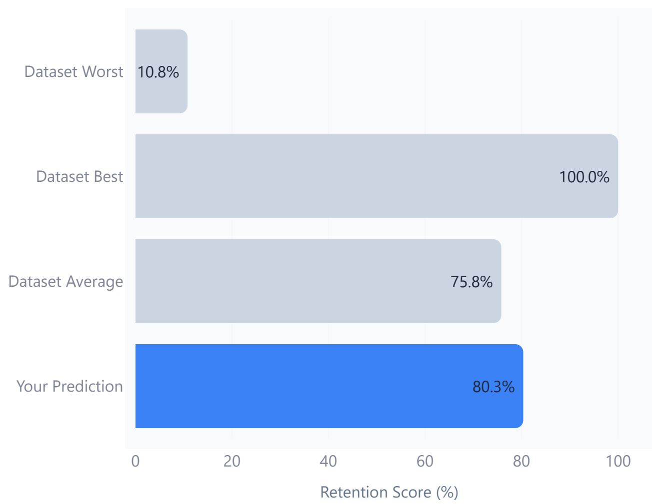
Your predicted retention score compared against the distribution of all students in the training dataset, clearly showing where you fall relative to peers.

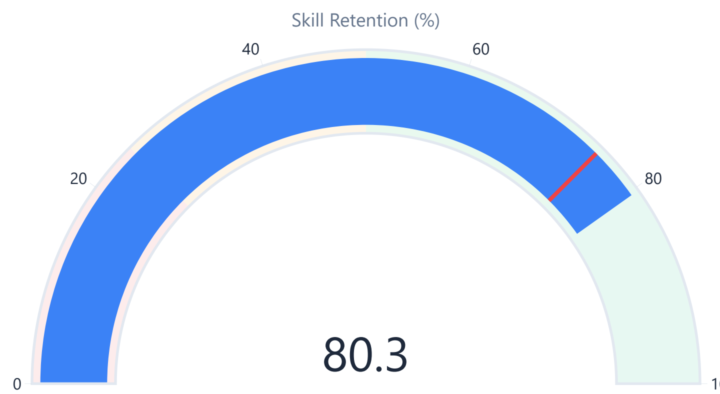
Performance vs Dataset

Your predicted retention score compared against the distribution of all students in our training dataset, showing where you fall relative to peers.

Retention Confidence Gauge

Skill retention score displayed on a 0–100 scale. Green zones (75–100) represent excellent long-term skill retention; red zones (0–25) indicate at-risk retention.





Retention Insights & Recommendations

Specific, actionable recommendations tailored to your learning profile to maximize skill retention and long-term mastery.

Retention Insights

Beginner-level prompt skills limit the effectiveness of AI-assisted learning. Poor prompts lead to weak outputs, wasting study time and confusing content mastery.

Lower pre-semester GPA suggests foundational gaps or inefficient study strategies. Addressing these will improve retention this semester.

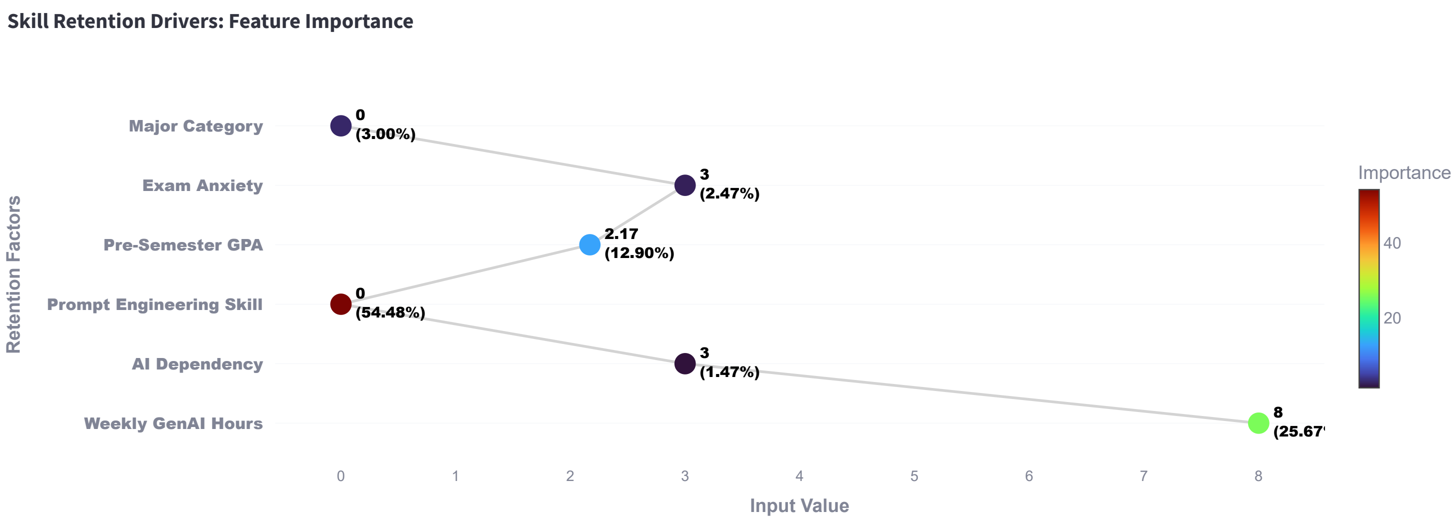
Recommendations

Invest 5–10 hours this semester in structured prompt engineering training. Platforms like Coursera offer free micro-courses; improved prompts directly improve retention.

Focus on foundational concepts before advanced topics. Use spaced repetition and active recall techniques—proven to improve long-term retention.

What Drives Skill Retention?

Analysis of factors that most strongly influence whether students retain foundational skills while using generative AI. Features are ranked by model importance—darker blue indicates stronger predictive power for skill retention outcomes.



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Key Insights & Recommendations

Critical Findings: The Data-Driven Reality

Non-Linear Burnout Escalation

AI usage hours reveal a counterintuitive pattern where burnout risk does not increase linearly instead, it plateaus safely in a narrow **"Optimal Window"** before accelerating into a crisis zone that negates all academic gains.

Window

- **5–10 hrs/week:** 19% high burnout | **Peak GPA (+0.203)**
- **10–15 hrs/week:** 34.1% high burnout | *Risk doubles*
- **25–40 hrs/week:** 83.1% high burnout | **GPA plummets back to baseline**

Dependency Crisis

High perceived AI dependency (6+/10) creates a catastrophic shift in outcomes; students exceeding this threshold experience a **65% surge in dependency metrics** coupled with **skill retention plummeting to 62%**—erasing all protective academic benefits gained from moderate tool usage.

At Extreme Dependency

- **94.7%** experience high burnout.
- **Skill retention:** 62% (vs. 76% at moderate use).
- **GPA advantage:** None—actually negative.

Trilemma: No Escape Without Balance

The data reveals a structural three-way tension where students cannot simultaneously optimize for heavy AI usage, low burnout, and high skill retention—this forces a deliberate trade-off where the only path to all three goals is **restrained, purposeful usage** complemented by **robust traditional study habits**.

The Protection Factor

- **Traditional study hours:** −0.136 correlation with burnout.
- **Weekly GenAI hours:** +0.469 correlation (strongest predictor).

What Actually Drives Student Outcomes? Statistical Evidence

Factor	Burnout Impact	GPA Importance	Actionability
Weekly GenAI Hours	+0.469 (Strongest)	0.90%	🟥 High Control
Pre-Semester GPA	-0.096 (Mild Protection)	95.12% (Dominant)	🟩 Historical Anchor
Traditional Study Hours	-0.136 (Strongest Shield)	2.89%	🟥 High Control
Perceived AI Dependency	+0.369 (High Risk)	0.26%	🟥 Observable Signal
Prompt Engineering Skill	N/A	0.36%	🟥 High Control
Anxiety Level During Exams	+0.165 (Stressor)	N/A	🟧 Moderate Control
Tool Diversity	-0.007 (Negligible)	0.14%	⬛ Irrelevant

Key Takeaway

The data unambiguously shows that *tool proliferation* (using 3–5 different AI platforms) has virtually no relationship with burnout or performance outcomes (**correlation = -0.007**), disproving the intuition that students are overwhelmed by software complexity. Instead, the sheer *volume of hours* spent offloading cognitive work (**r = +0.469**) is the singular dominant driver of both burnout risk and skill degradation, while *traditional offline study hours* serve as the strongest operational buffer against psychological strain.

The Hidden Multiplier: Advanced Prompt Engineering Transforms Outcomes

User Level	Avg GPA	Skill Retention	Retention Gain	Strategic Implication
Beginner	3.334	71.1	Baseline	AI as shortcut delivery system (stunts learning)
Intermediate	3.334	75.8	+4.7 pts (+6.6%)	AI as assistant (neutral learning environment)
Advanced	3.389	82.1	+11 pts (+15.5%)	AI as pedagogical framework (accelerates mastery)

The Insight

Investing in prompt engineering proficiency fundamentally reframes how students interact with generative AI transforming it from a *cognitive shortcut that erodes learning* into an *interactive tutoring system that enhances mastery*. Advanced users gain a **+11 point skill retention advantage** (82.1 vs. 71.1) with minimal additional GPA impact, suggesting that the real competitive edge comes from deeper learning rather than raw task automation.

Institutional Policy Paradox: Why Bans Backfire

Policy	Avg GPA	Skill Retention	High Burnout %	Dependency Score
Allowed with Citation	3.35	75.91	23.83%	3.51
Actively Encouraged	3.35	76.13	23.77%	3.52
Strict Ban	3.33	74.99	29.76%	3.48

The Evidence

1. **Strict Bans** ✗ No reduction in dependency.
2. **Strict Bans** ✗ No GPA improvement (+0.02 lower).
3. **Strict Bans** ✗ **25% burnout increase** (strongest negative outcome).
4. **Actively Encouraged Policy** ✓ Highest skill retention (76.13%).
5. **Allowed with Citation Policy** ✓ Best balance between burnout reduction and skill retention.

The Paradox

Implementing a **Strict Ban** does *not* reduce dependency metrics (remaining at 3.48—virtually identical to permissive policies), yet it **surges high burnout by 25% to 29.76%**. This counterintuitive result suggests that prohibition frameworks generate *anxiety and friction* around enforcement rather than productive learning environments, essentially punishing students without addressing root causes.

Executive Summary: 50,000-Student Dataset Key Metrics

Metric	Value	Strategic Implication
Students Improved GPA	87.52% (43,760 students)	Widespread positive outcomes overall
Avg GPA Improvement	+0.203 points (+2.6% baseline)	Meaningful but modest absolute gains
Optimal AI Usage Window	5-15 hours/week	Evidence-based usage ceiling
High Burnout at Optimal Use	19-34.1% (19% optimal)	Manageable risk level
High Burnout at Heavy Use	83.1% (at 25-40 hrs/week)	Unsustainable and counterproductive
Burnout Risk Multiplier	7.9× increase from low to high use	Usage volume matters enormously
Students at High Burnout Risk	24.97% (12,487 students)	Urgent mental health intervention needed
Advanced Users Retention Gain	+11 points (+15.5%) vs. beginners	Prompt engineering is a skill multiplier
Strongest Burnout Driver (Weekly GenAI Hours)	+0.469 correlation (r***)	Primary burnout control point
Strongest Protective Factor (Traditional Study Hours)	-0.136 correlation (strongest buffer)	Offline study is non-negotiable
[illegible]	[illegible]	[illegible]

The Bottom Line: A Three-Part Synthesis

1

Volume, Not Variety, Drives Outcomes

Using 3 AI tools vs. 5 tools has virtually zero impact (r = -0.007), whereas using AI for 5 hours vs. 25 hours per week determines whether students thrive or burn out. Policymakers and institutions have been fixating on the wrong variable; it's not *which* tools students use, it's *how much time* they spend on tool-mediated work without reciprocal offline processing.

2

The Skill Retention Multiplier is Real

Students who master prompt engineering and use AI for iterative reasoning (debugging, brainstorming, synthesis) see a **+11-point skill retention boost** compared to those who treat AI as a shortcut. This suggests that the competitive edge belongs to students who *learn to think differently with AI*, not those who merely use it more.

3

Prohibition Paradoxically Worsens Outcomes

Strict bans increase burnout by 25%, generate enforcement anxiety, and fail to reduce dependency. The evidence-based alternative; **"Allowed with Citation"** paired with explicit usage guidelines and digital literacy training; produces better outcomes across all measured dimensions (burnout, GPA, and skill retention).

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Key Insights & Recommendations

Key Insights | Stakeholder Recommendations

Evidence-Based Recommendations by Stakeholder

Select Stakeholder Group

- (●) Students
- ( ) Educators
- ( ) Institutions
- ( ) Policymakers

For Students: The Science-Backed Playbook

✓ DO: Operate in the Optimal Window (5–15 hours/week)

The empirical evidence is unambiguous—the peak academic ROI occurs at **5-10 hours/week** where students achieve GPA gains of **+0.203** with only 19% high burnout risk. Extending usage to 10-15 hours/week maintains solid outcomes but doubles burnout risk; exceeding 20 hours/week triggers cascading negative effects that eventually neutralize all academic gains by 25-40 hours/week (83.1% burnout, GPA back to baseline).

Action Items

- Set a weekly AI usage budget of **10–12 hours** (middle of optimal range).
- Track your weekly usage using built-in tool analytics.
- Schedule AI sessions in advance; avoid continuous/informal usage throughout the day.

✓ DO: Master High-Value Use Cases

The report identifies that AI is most effective for **debugging/troubleshooting** (24.59% adoption, high skill gain), **ideation/brainstorming** (21.44% adoption, creative leverage), and **summarizing/synthesizing** (17.27% adoption, time efficiency). These use cases build *scaffolding skills* that transfer to novel problems, whereas direct answer generation (12.68% adoption) demonstrates the weakest learning outcomes and highest dependency risk.

Action Items

- Use AI for **iterative problem-solving**, not direct answers.
- Treat AI outputs as *starting points for critical analysis*, not final submissions.
- Deliberately avoid asking AI to generate complete solutions or essays.

✓ DO: Develop Advanced Prompt Engineering Skills

Students with advanced prompt engineering proficiency demonstrate a **+11 point skill retention advantage** (82.1 vs. 71.1 for beginners) and perceive AI as a pedagogical partner rather than a cognitive crutch. This skill multiplier effect suggests that the competitive edge belongs to those who *master how to interact* with AI, not those who merely use it.

Action Items

- Invest 2–3 hours/week in prompt engineering courses or self-study.
- Practice specificity, step-by-step reasoning, roleplay framing and iterative refinement.
- Track how prompt quality directly impacts output value (your feedback loop).

✓ DO: Monitor Your Dependency Score and Burnout Signals

Students with AI dependency scores above 6/10 experience a **structural collapse** in skill retention (62%) and spike in burnout (94.7%). Early warning signs include relying on AI for routine tasks without trying independent problem-solving, feeling anxiety when AI is unavailable, and noticing that your performance on *non-AI tasks* has deteriorated.

Action Items

- Use the dashboard's AI Dependency tracker (self-reported 1-10 scale).
- If your score exceeds 5, intentionally increase non-AI study blocks (20-30 min uninterrupted work).
- Maintain offline study hours at 10+ hours/week as a burnout buffer.

✗ DON'T: Exceed 20 Hours/Week (The Diminishing Returns Cliff)

Beyond 20 hours/week, every additional hour of AI usage correlates with increased burnout and decreased skill retention, with negligible GPA gains. The data shows a non-linear acceleration where 25-40 hours/week produces an 83.1% burnout rate and actually *reduces* GPA back to baseline (3.2).

✗ DON'T: Rely on "Direct Answer Generation" (The Skill Killer)

Using AI to generate complete answers, full essay drafts, or homework submissions without critical engagement produces the lowest skill retention outcomes and highest dependency. The cognitive trade-off is that you trade short-term task completion for long-term learning degradation.

✗ DON'T: Ignore Burnout Signals or Skip Traditional Study

The strongest protective factor against burnout is *traditional offline study hours* ($r = -0.136$), which serves as a more powerful buffer than reducing AI usage itself. Paradoxically, some students attempt to substitute AI for structured thinking time, which backfires dramatically.

Key Insights & Recommendations

Evidence-Based Recommendations by Stakeholder

Select Stakeholder Group

○ Students ● Educators ○ Institutions ○ Policymakers

For Educators: Design for Mastery, Not Automation

✓ DO: Teach Explicitly Which AI Use Cases Are Beneficial

Rather than banning or ignoring AI, the evidence strongly supports explicit instruction on *how and when* to use these tools productively.Students need scaffolding to understand that debugging, ideation, and summarization are high-value use cases, while direct answer generation is a skill trap. Your teaching becomes a *guardrail against dependency* rather than prohibition.

Curriculum Integration

- Embed lessons on "responsible AI workflows" in Week 2-3 of your course
- Show examples of low-value vs. high-value prompts with learning outcome data
- Assign reflection exercises where students evaluate their own AI usage effectiveness

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✓ DO: Model Responsible AI Use Yourself

Educators who demonstrate thoughtful AI integration (using it for brainstorming lessons, not replacing grading; leveraging it for creating diverse practice problems, not automating feedback) set a powerful cultural norm. Students internalize your AI literacy model.

✓ DO: Assess for Deep Learning, Not Answer Correctness

Shift your assessment strategy to evaluate *reasoning, transfer, and problem-solving process* rather than just final answers. This automatically disadvantages students who rely on AI for direct answers while rewarding those who use it as a thinking tool. The format change (move from multiple-choice/short-answer to open-ended written justifications) raises the bar against pure automation.

Assessment Redesign

- Include "explain your reasoning" and "why did you choose this approach" components
- Use Socratic follow-up questions orally to verify understanding depth
- Weight problem-solving process and intermediate steps equally with final answers

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✓ DO: Support Students Showing AI Dependency Signals

Watch for students who:

1. Submit work that lacks personal analysis or voice.
2. Struggle with follow-up questions about their own submissions.
3. Show anxiety around in-class or timed assessments.
4. Report using AI 20+ hours/week.

These are high-risk students who benefit from *structured guidance* rather than judgment.

Intervention Strategy

- Offer optional office hours focused on "healthy AI workflows"
- Recommend the prompt engineering skills pathway
- Connect students to campus mental health resources for burnout

✓ DO: Design Assignments AI Cannot Fully Solve

The most effective pedagogical approach is to *structurally require* human judgment and creativity. This isn't prohibition—it's intentional course design.

Assignment Design Principles

- Include AI-free assessments (closed-book exams, in-class problem-solving)
- Create assignments that require applying novel concepts to *unique, personalized scenarios* (AI struggles with true novelty)
- Emphasize transfer to domains the model has never seen
- Use "show your work" requirements that expose where AI was or wasn't appropriate

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Key Insights & Recommendations

Evidence-Based Recommendations by Stakeholder

Select Stakeholder Group

○ Students ○ Educators ● Institutions ○ Policymakers

For Institutions: Policy as Intervention, Not Prohibition

✓ DO: Adopt the "Allowed with Citation" Policy Framework

The empirical evidence is decisive: the **"Allowed with Citation"** policy yields the best overall outcomes across burnout (23.83%), skill retention (75.91%), and GPA (3.35), while avoiding the backfire effects of prohibition. This policy acknowledges AI's reality while creating transparency and accountability.

Policy Language

- "AI tools are permitted in this course with explicit disclosure of where they were used and for what purpose"
- "Citations should note which tool was used (e.g., ChatGPT, Claude, etc.) and the specific task"
- "Grading rubrics reward thoughtful AI integration, not mere automation"

✗ DO NOT: Implement Strict Bans (They Backfire)

The data is unambiguous—strict bans **increase high burnout by 25% to 29.76%** without reducing dependency or improving outcomes. Bans create anxiety, enforcement friction, and underground behavior while doing nothing to address root causes. A ban essentially says "this tool exists and is powerful, but you can't learn to use it responsibly here"; a message that contradicts good pedagogy.

Why Bans Fail

- No reduction in dependency (3.48 under ban vs. 3.51-3.52 under permissive policies)
- Highest burnout outcome (29.76% vs. 23.8% with citation policies)
- Creates "shadow usage" where students use AI secretly without guidance

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✓ DO: Provide Clear Usage Guidelines (10-15 hours/week)

Institutional communications should explicitly recommend the **optimal usage window** (5-15 hours/week) with supporting data. This shifts the narrative from "are you using AI?" to "are you using it wisely?"

Guidance Framework

- **Optimal Zone (5-15 hrs/week):** Sweet spot for GPA gains with manageable burnout
- **Caution Zone (15-25 hrs/week):** Burnout risk rising (55.8% high burnout at 25 hrs)
- **Crisis Zone (25+ hrs/week):** Severe burnout (83.1%) and skill degradation; academic gains reversed

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✓ DO: Invest in Digital Literacy & Prompt Engineering Programs

The data shows a **+11 point skill retention multiplier** for advanced prompt engineering users. Offering institution-wide workshops on "AI literacy" and "prompt engineering fundamentals" is a direct, high-ROI intervention that shifts students toward the mastery pathway.

Program Components

- **Weeks 1-2:** Foundations (what is an LLM, limitations, hallucination risks)
- **Weeks 3-5:** Prompt engineering fundamentals (specificity, iteration, step-by-step reasoning)
- **Weeks 6-8:** Discipline-specific applications (debugging for CS, research synthesis for humanities, etc.)
- Mandatory attendance for students with dependency scores 6+ or burnout signals

✓ DO: Ensure Universal Access to AI Tools

Currently, **42.3% of students pay for premium tiers** while 57.7% use free versions. Ensure equitable access by offering institutional licenses or credits to all students, preventing a two-tier system where wealthier students have better tools and outcomes. This is an equity issue with measurable impact.

✓ DO: Create Mental Health & Burnout Support Infrastructure

With 25% of students in high burnout risk and a direct correlation between AI usage intensity and psychological strain, institutions should proactively integrate AI-usage discussions into campus mental health protocols.

Support Structure

- Train counselors on recognizing AI dependency as a burnout driver
- Offer workshops on "healthy technology boundaries" or "digital wellness"
- Create drop-in "AI usage review" sessions where students get feedback on their workflows

Dataset Overview & Exploration | Advanced Analytics | Machine Learning Models | Predictions | Insights

Key Insights & Recommendations

Key Insights | Stakeholder Recommendations

Evidence-Based Recommendations by Stakeholder

Select Stakeholder Group

- ○ Students
- ○ Educators
- ○ Institutions
- ◉ Policymakers

For Policymakers: Building Sustainable AI Literacy Infrastructure

✓ Prioritize: Permit with Responsibility (Not Prohibition)

The national evidence from 50,000 students is clear; banning AI in education increases student burnout, creates enforcement friction, and does nothing to reduce dependency. Instead, policymakers should mandate that institutions adopt **"Allowed with Citation"** frameworks paired with concrete usage guidelines. The policy shift costs nothing and measurably improves outcomes across burnout, skill retention, and academic performance.

Policy Recommendation

- State/national policy should *permit* AI in educational settings with institutional citation/disclosure requirements.
- Guidance should emphasize the **5-15 hour/week optimal window** with burnout risk data.
- Policy should explicitly reject blanket bans as evidence-contraindicated.

✓ Prioritize: Fund Digital Literacy as Foundational Infrastructure

Just as literacy programs are infrastructure investments, digital literacy and prompt engineering proficiency must become foundational competencies. The 15% skill retention advantage for advanced users (82.1 vs. 71.1) suggests that students who learn *how to think with AI* gain a durable competitive edge.

Funding Mechanisms

- Direct grants to school districts/universities for AI literacy curriculum development.
- Teacher training programs on integrating AI responsibly into pedagogy.
- Creation of open-source prompt engineering curricula (eliminate vendor lock-in).
- Scholarships for students to complete certification programs.

✓ Prioritize: Ensure Equitable Access (No Cost Barriers)

With 42.3% of students paying for premium AI access, there is a clear stratification where wealthy students have superior tools. National or state policies should ensure that all students have access to at least one high-quality AI tool via institutional subscriptions or subsidies. This is a *civil rights issue* framed as a technology policy.

Policy Framework

- Require institutions receiving public funding to provide universal AI tool access.
- Subsidize or negotiate group licensing for affordable access to premium models.
- Prevent vendor monopolies by requiring multi-tool institutional strategies.

✓ Prioritize: Support Longitudinal Research on AI's Long-Term Educational Impact

This dataset captures cross-sectional outcomes in one academic term. Policymakers should fund multi-year longitudinal studies tracking students' AI usage patterns, skill development, career outcomes, and earnings trajectories. Do advanced users earn more? Do they retain skills longer in workplace settings? These questions require years of data.

Research Agenda

- Fund 5-year longitudinal studies tracking cohorts from first year through post-graduation.

- Collect data on workplace skill transfer (does high skill retention translate to job performance?).
- Compare lifetime earnings of high-skill vs. high-dependency graduates.
- Track psychological outcomes and burnout resolution post-graduation.

✗ Avoid: One-Size-Fits-All Policies

Different disciplines, student populations, and institution types have different optimal AI integration strategies. Computer science has different pedagogy than literature; students with STEM backgrounds differ from humanities students. Policy frameworks should be *flexible blueprints*, not rigid mandates.

Anti-Pattern

- ✗ "All AI use is prohibited institution-wide"
- ✗ "All courses must use AI for every assignment"
- ✓ "Institutions must adopt a usage framework that meets these evidence-based criteria"

✗ Avoid: Ignoring Equity Issues

The data shows a clear bifurcation between students with premium tool access and free-tier users. Without intentional intervention, AI's productivity gains will accrue only to wealthy students, widening the achievement gap. This is not a technology problem, it's a *policy choice*.
